"Developing a Governance-Led Decision Support Framework to Strengthen Cross-Border Financial Oversight at DMCR."

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Declaration

Declaration I: Declaration of Own Work

I, Mirza Muhammad Wali Baig (Student ID: 24125373), hereby declare that this Business Transformation Project, submitted in partial fulfilment of the requirements for the degree of MSc in Project Management at Arden University, is my own original work. It has not been submitted, in whole or in part, for any other degree or qualification at this or any other institution. All sources used have been acknowledged and properly referenced in accordance with Arden University Harvard referencing conventions.

Name: Mirza Muhammad Wali Baig **Date: 8th April 2026**

Signature: MirzaM.WaliBaig

Declaration II: Declaration of non-confidentiality

I declare that the work contained in this Business Transformation Project is not confidential and does not contain commercially sensitive, legally protected, or personally identifiable information that would prevent it from being shared, stored, or used for academic purposes. All organizational data from DM Clinical Research (DMCR) has been treated at the governance-structure level only, anonymized where appropriate, and no proprietary information has been reproduced in a form that could cause harm to any individual or organization.

Name: Mirza Muhammad Wali Baig **Date: 8th April 2026**

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Name: Mirza Muhammad Wali Baig **Date: 8th April 2026**

Signature: MirzaM.WaliBaig

Acknowledgements

I begin by expressing my deepest gratitude to Almighty Allah, whose blessings, guidance, and mercy granted me the strength, perseverance, and resources to accomplish this work. Without His will, none of this would have been possible.

I would like to extend my sincere appreciation to my supervisor, Ahmad Al-Kaakaty, for his invaluable guidance, thoughtful feedback, and continuous encouragement throughout this research journey. His patience in reviewing multiple drafts and his commitment to academic rigor significantly enhanced the quality and clarity of this work.

I am also grateful to the management and faculty of Arden University for providing a supportive academic environment and a practice-oriented learning framework. The program's emphasis on applied research has been instrumental in shaping this study and enabling its real-world relevance.

I would like to acknowledge DM Clinical Research (DMCR), whose operational context inspired this research. The organizational challenges examined in this study provided a meaningful foundation for the development of the proposed governance framework. This work has been conducted with full respect for confidentiality, and it is my hope that the outcomes will contribute positively to the organization's continued growth and improvement.

Finally, and most importantly, I offer my heartfelt gratitude to my family, my parents, sisters, and close relatives, whose unwavering support, sacrifices, and belief in me have been my greatest source of strength. Their encouragement sustained me through the challenges of balancing professional responsibilities with academic commitments, and this achievement is as much theirs as it is mine.

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Abstract

The Globalization of business operations has created governance challenges that domestically structured organizations do not face. Where clinical trial operations and financial administration are separated across national borders, divergent regulatory environments, and time-zone differentials, the mechanisms through which accountability is allocated, decisions are made, and compliance is assured become structurally vulnerable to fragmentation. This Study addresses that challenge at DM Clinical Research (DMCR), a Houston, Texas-based multi-therapeutic clinical trial network whose financial administration function is managed from Pakistan, which creates bifurcation in its governance environment, characterized by four compounding deficiencies: the absence of formally defined decision rights, undefined escalation pathways, weak finical oversight with regards to approval controls, and structural communication failures between the two functions.

The study adopts a Design Science Research (DSR) methodology, operationalized through the six-stage DSRM process model of Peffers, et al. (2007) within a qualitative single-case study design. All data are derived exclusively from secondary documentary sources, including DMCR's internal governance documentation, financial reconciliation records, interdepartmental communication logs, and publicly available regulatory standards. This approach ensures full compliance with the ethical approval granted under Project Reference P18478. An interpretivist philosophy and abductive reasoning approach underpin the analytical logic, enabling iterative movement between established governance theory and DMCR's documentary evidence.

The five EDM domain goals of COBIT 2019 and all six principles of ISO/IEC 38500:2015 are used to compare DMCR's governance arrangements. The findings reveal that DMCR operates predominantly Level 0 to Level 1 maturity across all five EDM objectives. Significantly below the Level 3 (Established) threshold that (ISACA, 2019) identifies as the minimum capability level for organizations of comparable complexity. Four of the six ISO 38500 principles show non-aligned or partially aligned status. Critically, the analysis sets out that these deficiencies are structural in origin, attributable to the absence of governance architecture rather than individual performance failures.

In response, the study designs a governance-led decision support framework comprising six interrelated artefacts: a Cross-Border RACI Decision Rights Matrix; Financial Oversight Control Layers with three approval tiers; Accountability Allocation Structures; time-bound Escalation Protocols; Governance Review Checkpoints at monthly, quarterly, and annual cadences; and Reporting Transparency Mechanisms incorporating defined KPIs and Power BI-supported audit trail infrastructure. The framework is evaluated against both COBIT 2019 and ISO/IEC 38500:2015, confirming full coverage of all five EDM domain objectives and all six governance principles. A phased implementation of roadmap deploys the artefacts across three sequential phases over six months.

Cross-sector validation through analogous governance cases from HSBC, Microsoft, Toyota, and Unilever, alongside sector-level benchmarking from the (SCRS, 2025) and the global action plan for strengthening clinical trial ecosystem by world health organization report (WHO, 2025), confirms that the governance failures identified at DMCR are structurally representative of the clinical research sector rather than organizationally exceptional case study. The framework's implementation is projected to advance DMCR's governance maturity from its current Level 0–1 baseline to the Level 3 (Established) threshold across the COBIT 2019 EDM domain. Beyond DMCR, the study produces a replicable, standards-grounded governance design methodology applicable to comparable cross-border clinical research organizations.

Keywords: *Corporate governance; cross-border financial oversight; Design Science Research; COBIT 2019; clinical research site management*

Chapter 1 Introduction

1.1 Background

The globalization of business operations has fundamentally transformed the governance landscape for organizations functioning across multiple jurisdictions. As enterprises expand across geographic and regulatory boundaries, they face an intensifying challenge: ensuring that financial accountability, decision-making authority, and oversight mechanisms remain coherent despite the structural fragmentation that cross-border operations inherently produce (Aguilera & Ruiz, 2025); (Rituraj , et al., 2024). This Challenge is compounded by the absence of unified governance framework capable of harmonizing divergent regulatory environments, time-zone differentials, and culturally distinct operational practices within a single organizational structure (Carvalho & Shanaia, 2025). Governance design for globally dispersed organizations has consequently emerged as a critical area in contemporary project management and corporate governance scholarship (Bakhshi, et al., 2025).

1.2 The Clinical Research Sector and Cross-Border Governance

The clinical research sector operates under a uniquely stringent governance environment. Organizations conducting trials must comply with ICH-GCP guidelines, applicable national regulatory requirements including those of the U.S. Food and Drug Administration (FDA), and the financial reporting obligations of sponsor and contract research organization partners (Medidata and Society for Clinical Research Sites, 2017). Financial governance in this sector is therefore not merely an administrative function; it is a regulatory compliance obligation with direct bearing on trial data validity, sponsor trust, and organizational licensure. Where clinical operations and financial administration are separated across national borders, maintaining governance coherence intensifies considerably. Cross-border financial governance must simultaneously satisfy U.S. regulatory frameworks, international payment compliance requirements, and the operational demands of multi-site payment cycles — making governance a strategic operational imperative (SCRS and Greenphire, 2017) These pressures are directly operative in the case of DM Clinical Research (DMCR), whose cross-border clinical-finance structure transforms a general governance challenge into a specific, evidenced, and actionable research problem.

1.3 Organizational and Problem Context

DM Clinical Research (DMCR) is a Houston, Texas-based multi-therapeutic clinical trial network established in 2006, comprising 30 dedicated research sites and more than 700 employees conducting Phase I through Phase IV trials across multiple therapeutic areas including vaccines, internal medicine, pediatrics, gastroenterology, psychiatry, neurology, and women's health (DM Clinical Research, 2026).

Despite its operational scale, DMCR's financial administration function, encompassing accounts receivable, financial reconciliation, and cross-border payment processing; is administered from Pakistan, creating a structurally bifurcated governance environment in which clinical operations and financial oversight are separated by geography, regulatory jurisdiction, time-zone differential, and institutional culture. Secondary documentary analysis of DMCR's internal records, financial reconciliation artefacts, and inter-departmental communication logs reveals four compounding governance deficiencies: the absence of formally defined decision rights across clinical and financial functions; undefined escalation pathways for financial discrepancies; weak financial oversight and approval controls; and structural communication failures between the two functions. As the (Harvard Law School Forum, 2022)observes, oversight mechanisms do not scale automatically as operations expand — a condition directly evidenced in DMCR's current state. McKinsey and Company, (2022) Report further establish that approximately 80 percent of organizations report significant decision-making difficulties precisely where formal authority allocation frameworks are absent.

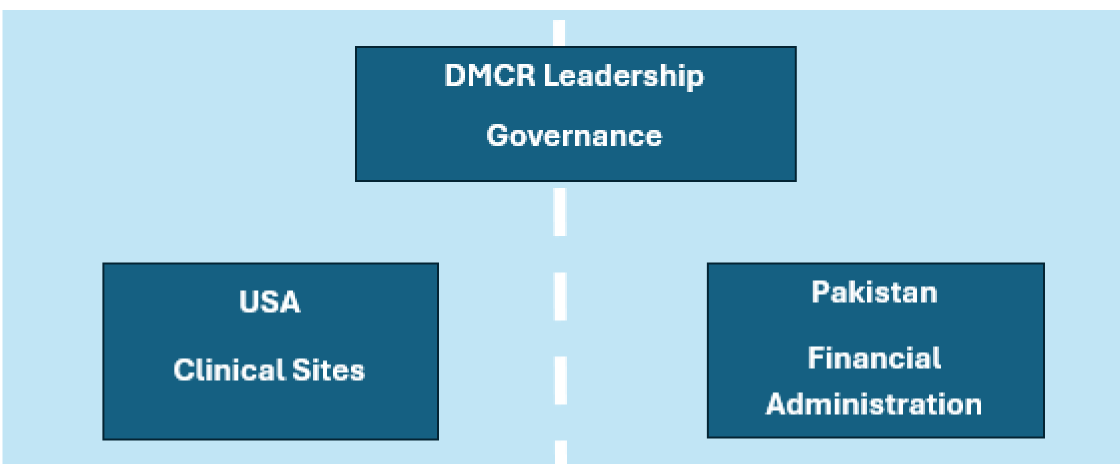

Figure 1.1: Visual representation of DMCR's Leadership's bifurcated Governance. (Personal collection, 2026)

1.4 Study's Significance

In response to the governance deficiencies identified above, this study pursues the following overarching aim and four supporting objectives, presented in Table 1.1.

RESEARH AIM	
Aim	To design a governance-led decision support framework that addresses DMCR's documented cross-border governance deficiencies and strengthens financial oversight through the application of COBIT 2019 and ISO/IEC 38500:2015 governance standards, using Design Science Research methodology.
RESEARCH OBJECTIVES	
O1	Analyze DMCR's existing governance and financial oversight arrangements through qualitative thematic analysis of secondary organizational documentation.
O2	Benchmark DMCR's current governance practices against COBIT 2019 EDM domain objectives and all six principles of ISO/IEC 38500:2015.
O3	Identify specific governance gaps in decision rights, accountability allocation, escalation protocols, and financial control mechanisms.
O4	Design a governance-led decision support framework comprising six deployable governance artefacts addressing the identified deficiencies.

Table 1.1: Research Aim and Objectives.

As stated in the objective, this study contributes to governance practice in the clinical research sector by producing a replicable, standards-grounded artefact set for cross-border organizations facing analogous deficiencies. Industry benchmarking confirms that DMCR's challenges are sector-representative: the SCRS Global Site Landscape Survey (2025) documents that 80 percent of over 12,000 sites surveyed report payment governance failures and absent financial oversight structures. The WHO's GAP-CTS (2025) explicitly identifies weak governance architecture as the systemic root cause of clinical research operational failures globally.

1.5 Scope and Limitation

This study focuses on DMCR's cross-border governance arrangements between its U.S. clinical operations and Pakistan-based finance function. Cloud-based analytics tools are treated as governance visibility infrastructure that supports, but does not substitute for, the formal governance controls designed in Chapter 5. The study relies exclusively on secondary documentary data; no primary data collection through interviews or surveys was undertaken, consistent with the approved ethics protocol under project reference P18478 as stated in Appendix 2.

1.6 Structure of the Report

Chapter 2 reviews the literature underpinning the study's theoretical framework. Chapter 3 presents the research methodology. Chapter 4 analyses DMCR's governance deficiencies through COBIT 2019 and ISO 38500 benchmarking. Chapter 5 designs and evaluates the governance framework. Chapter 6 provides a personal reflection on the research process and professional learning outcomes.

Chapter 2 Background Research and Literature Review.

2.1 Project Governance Theory: Evolution and Critical Evaluation

Project governance scholarship has evolved from early conceptualizations of governance as a project-level control mechanism towards multi-tiered models addressing accountability, decision rights, and performance measurement across entire organizational portfolios (Bakhshi et al., 2025; Brunet and Choinière, 2025). Turner and Müller (2003) establish the foundational distinction between project management — the operational execution of activities — and project governance, which concerns the authority structures and accountability frameworks through which projects are directed and controlled. This distinction is foundational for the present study because DMCR's problem is not one of project management inadequacy; it is one of governance architecture failure — a diagnostic that only becomes visible when governance is understood as structurally separate from management execution.

Subsequent scholarship both deepens and qualifies this foundation. Müller et al. (2023), Sankaran et al. (2025), and Zaman et al. (2022) share a core analytical conclusion: structural deficiencies in governance architecture, not individual behavioral failures, are the primary driver of governance breakdown. Where these scholars diverge is in their prescriptive emphasis. Sankaran et al. (2025) focus on the evolutionary trajectory of governance frameworks as organizations mature, while Zaman et al. (2022) provide a more operationally specific finding — that governance improvement is most effective when decision rights, escalation protocols, and high-performance working practices are introduced simultaneously rather than sequentially. For this study, Zaman et al.'s (2022) findings are particularly relevant because it directly justifies the integrated, multi-artefact design logic of Chapter 5: introducing a RACI matrix in isolation, without accompanying escalation protocols and accountability structures, would be analytically insufficient.

However, a critical limitation runs through this entire body of literature: it retains an intra-organizational orientation. Musawir et al. (2020) demonstrate that project governance contributes most significantly to strategy implementation when formally codified and role-specific, but their systematic review does not address governance operating across national regulatory boundaries. This is not a minor methodological gap; for DMCR, the defining governance challenge is precisely the absence of a framework capable of allocating authority across two functions separated by geography, regulatory jurisdiction, time-zone differential, and institutional culture. The project governance literature, however sophisticated its structural models, does not speak to this condition — a limitation that motivates the turn to cross-border governance theory in Section 2.2.

2.2 Cross-Border Governance Challenges: DMCR Context and Theory

Cross-border governance scholarship has established theoretical frameworks for understanding headquarters–subsidiary accountability dynamics and the challenge of maintaining governance coherence across divergent national regulatory environments (Wei et al., 2020; Rituraj et al., 2024). These two bodies of work broadly agree that cross-border governance failures are structurally produced — arising from institutional duality, regulatory divergence, and the absence of unified oversight mechanisms — rather than from individual managerial inadequacy. Wang et al. (2025) extend this consensus empirically, demonstrating that inter-organizational governance in geographically distributed environments is characterized by amplified accountability gaps, unresolved discrepancy accumulation, and systematic information asymmetry — conditions precisely documented in DMCR's operational records.

Where cross-border governance theory is less developed is in its prescriptive dimension. Wei et al. (2020) and Rituraj et al. (2024) produce analytical frameworks for understanding accountability dynamics; they do not produce deployable operational artefacts. This matters for the present study because understanding the problem is insufficient — DMCR requires a governance intervention, not merely a governance diagnosis. Project governance literature, conversely, designs artefacts but assumes a single-jurisdiction context. Neither body of literature, considered in isolation, is adequate for DMCR's governance challenge. The contribution of this study lies precisely at their intersection: applying DSR artefact design methodology to a cross-border governance diagnosis. Galorath (2025) provides empirical support for this intervention approach, establishing that formalized governance frameworks directly reduce operational failure rates in complex, distributed organizations — though his analysis, like the broader cross-border governance literature, does not extend to the clinical research sector specifically, leaving a gap that this study addresses directly.

2.3 COBIT 2019: Framework Strengths and Applicability

COBIT 2019 (ISACA, 2019) is organized across five domains encompassing 40 governance and management objectives. Its primary strength is the structural specificity with which it defines governance responsibilities across organizational roles, and its five-level capability maturity model — from Level 0 (Incomplete) to Level 5 (Optimizing) which enables organizations to measure current governance state against an externally recognized standard. The EDM domain — governing framework setting, benefits delivery, risk optimization, resource optimization, and stakeholder engagement — is of direct relevance to this study. Toifur et al. (2022) position EDM as formally embodying the governance principles of ISO/IEC 38500, creating a conceptual bridge between the two frameworks that justifies their combined deployment here. Balcon (2025) identifies COBIT's particular suitability for regulated healthcare and financial services contexts — both of which characterize DMCR's operating environment simultaneously, making COBIT the more operationally specific of the two frameworks selected.

However, COBIT 2019 carries a critical limitation that must be acknowledged: its governance objectives are defined at a level of abstraction that requires contextual customization before they can govern actual operational behavior (Chaudhuri, 2011). A framework that specifies that decision rights should be formally allocated does not specify how those rights should be structured for a U.S.–Pakistan cross-border clinical research network operating under dual regulatory obligations. This is not a weakness unique to COBIT; it is an inherent feature of any framework designed for broad applicability. The analytical implication is that COBIT 2019 functions as a diagnostic and evaluative standard in this study — not as a ready-made solution. The customization gap is precisely what Chapter 5 addresses, translating COBIT's EDM objectives into six DMCR-specific, deployable governance artefacts grounded in the specific conditions identified in Chapter 4.

2.4 ISO/IEC 38500 and Comparative Framework Analysis

ISO/IEC 38500:2015 provides a principles-based governance framework organized around six governing principles: Responsibility, Strategy, Acquisition, Performance, Conformance, and Human behaviors (ISO/IEC, 2015). Where COBIT 2019 is prescriptive and operational — assigning specific objectives to defined roles with a measurable maturity model — ISO 38500 is evaluative and normative, providing criteria against which governance decisions are assessed without prescribing how compliance is achieved. CIO Index (2025) characterizes this distinction precisely: COBIT offers structure and control; ISO 38500 offers principles and posture.

This difference in orientation is not a deficiency in either framework but rather a complementarity that makes their combined deployment analytically superior to deploying either alone. COBIT 2019 alone would produce well-structured governance artefacts without a principled governance rationale capable of operating across jurisdictions; ISO 38500 alone would articulate the right governance principles without the operational specificity required to produce deployable artefacts. For DMCR specifically, ISO 38500's jurisdiction-neutral design is a strategic advantage: it offers adaptability across both U.S. FDA governance requirements and Pakistani financial regulatory norms — a flexibility that COBIT 2019's more prescriptive architecture does not provide to the same degree. Visitsilp and Bhumpenpein (2021) empirically confirm this complementarity, demonstrating that deploying both frameworks together produces more comprehensive governance coverage than either standard independently. Aflakhah and Soewito (2024) validate this finding in a geographically distributed, multi-site organization structurally comparable to DMCR — providing the closest available external precedent for the dual-framework methodology adopted in this study.

The two ISO 38500 principles of greatest relevance to DMCR are Responsibility and Conformance. Chapter 4 establishes that both are structurally absent from DMCR's current governance arrangements: no RACI-based authority allocation exists, and no compliance monitoring mechanism governing cross-border payment processing has been documented. This means ISO 38500 functions in this study both as a normative benchmark against which deficiencies are measured and as an evaluative standard against which the designed framework is ultimately assessed — a dual role that justifies its selection alongside COBIT 2019.

2.5 Decision Rights, Accountability, and Escalation Frameworks

Decision rights — the formal allocation of authority to make specific categories of decision — represent a foundational governance mechanism whose absence has been empirically linked to compounding financial and operational failure in cross-border organizations. De Smet et al. (2022) document that approximately 80 percent of companies report decision-making difficulties, with the most persistent failure modes being accountability diffusion, escalation avoidance, and the absence of formally defined authority boundaries. Wang et al. (2025) extend this analysis specifically to cross-border governance contexts, demonstrating that undefined escalation pathways amplify accountability gaps and produce systematic information asymmetry. The agreement between De Smet et al. (2022) and Wang et al. (2025) on the primacy of structural authority allocation is significant: it confirms that accountability diffusion is not a sector-specific or organization-specific phenomenon but a structural governance failure mode that manifests wherever formal decision rights are absent — including, as Chapter 4 demonstrates, at DMCR.

Sari (2023) provides complementary evidence from comparable organizational settings, establishing that governance improvement is most significant when decision-control frameworks, independent oversight mechanisms, and formal accountability ownership are introduced simultaneously rather than in isolation. This finding has a direct design implication for Chapter 5: it rules out a minimalist intervention — such as introducing only the RACI matrix — in favor of the integrated, six-artefact framework architecture. The three strands of literature examined here provide convergent and mutually reinforcing prescriptions, which strengthens the evidence base for the integrated design logic of Chapter 5 rather than leaving it dependent on a single source.

2.6 Design Science Research (DSR) Methodology

Design Science Research was formalized by Hevner et al. (2004) as a paradigm generating prescriptive knowledge through the iterative design, development, and evaluation of artefacts intended to resolve identified organizational problems. Its distinguishing characteristics and the primary justification for its selection over descriptive alternatives in this study — is its explicit dual orientation towards academic rigor and practical relevance. A purely qualitative or interpretive approach would have produced an analytical account of DMCR's governance failures; DSR demands that the research go further and produce a deployable solution, evaluated against independently verifiable standards. Peffers et al. (2007) operationalized this through a six-stage DSRM process model, which this study adopts as its procedural framework. Baskerville et al. (2018) confirms that standards-based evaluation — assessing an artefact against established external standards rather than through empirical deployment — constitutes a methodologically rigorous DSR evaluation strategy, directly applicable to this study where deployment testing was precluded by the secondary-data design. Sengik et al. (2022) provides the most directly relevant precedent, producing an IT governance model for a structurally complex, multi-unit institution using an identical secondary-data, standards-validation DSR methodology — confirming that this approach is not methodologically novel but is consistent with established academic practice in governance framework design research.

2.7 Critical Synthesis and Identified Research Gap

This review reveals a literature characterized by partial but complementary coverage of the governance problem this study addresses. Three analytical conclusions define the research gap with precision.

First, what the literature already covers: project governance scholarship has produced sophisticated structural models of governance failure and remediation (Sankaran et al., 2025; Zaman et al., 2022); cross-border governance research has established theoretical accountability frameworks for distributed multi-jurisdictional organizations (Wei et al., 2020; Rituraj et al., 2024); COBIT 2019 and ISO 38500 have been validated as complementary governance standards with empirical effectiveness in complex, distributed settings (Visitsilp and Bhumpenpein, 2021; Aflakhah and Soewito, 2024); and DSR has been established as a rigorous methodology for producing deployable governance artefacts in institutional contexts (Sengik et al., 2022; Hevner et al., 2004).

Second, what remains insufficiently addressed: no existing body of research integrates these four elements — project governance theory, cross-border accountability frameworks, dual-standard COBIT 2019 and ISO 38500 benchmarking, and DSR artefact design methodology — into a prescriptive, deployable framework specifically designed for cross-border clinical research organizations in which clinical operations and financial administration are institutionally and jurisdictionally separated. The shared limitation of the project governance and cross-border governance literatures is that neither produces deployable governance artefacts for this specific organizational configuration. The shared limitation of the COBIT 2019 and ISO 38500 frameworks is that their combined application to clinical-financial cross-border governance settings has not been documented in the published literature.

Third, how this study responds: by applying DSR methodology to design a context-specific, standards-grounded governance framework comprising six deployable artefacts calibrated to DMCR's U.S.–Pakistan operational structure, this project fills a documented gap at the intersection of cross-border governance theory, clinical research governance practice, and standards-based artefact design. The contribution is dual: an immediately deployable organizational solution for DMCR, and a replicable governance design methodology for comparable cross-border clinical research organizations facing analogous structural deficiencies.

Chapter 3 Research Question and Methodology

3.1 Research Questions

The study's primary aim, four supporting objectives, and four research questions are presented in table 3.1. This structured presentation makes it explicit for understanding the direct logical chain connecting DMCR's identified governance problem to the designed framework artefact, in accordance with the DSR problem-to-solution traceability requirement of Hevner, et al. (2004).

RESEARH AIM	
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RESEARCH OBJECTIVES	
O1	Analyze DMCR's governance and financial oversight arrangements through qualitative thematic analysis of secondary organizational documentation.
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O3	Identify specific governance gaps in decision rights, accountability allocation, escalation protocols, and financial control mechanisms.
O4	Design a governance-led decision support framework comprising six deployable governance artefacts addressing the identified deficiencies.
RESEARCH QUESTIONS	
RQ1	What governance deficiencies contribute to DMCR's recurring financial discrepancies and cross-border communication failures, as evidenced through secondary organizational documentation?
RQ2	How do DMCR's current governance practices align with COBIT 2019 EDM domain objectives and ISO/IEC 38500:2015, and at what maturity level does the organization operate?
RQ3	What governance structure — comprising formal decision rights, accountability allocation, escalation protocols, and financial oversight controls — can improve cross-border financial oversight and decision transparency at DMCR?
RQ4	How can cloud-based analytics tools be integrated as governance visibility infrastructure to support, without substituting for, the governance controls embedded in the designed framework?

Table 3.1: Research aim and objectives and Research Objectives

3.2 Research Philosophy and Approach, and Methodological Logic

This study is grounded in an interpretivist research philosophy, which holds that organizational phenomena are socially and institutionally constructed and are best understood through the interpretive analysis of contextual documentary evidence Saunders et al. (2019). Interpretivism is the appropriate epistemological starting point because DMCR's governance deficiencies are embedded in the specific regulatory, geographic, and institutional conditions of the U.S.–Pakistan operational structure and require contextual interpretation to be meaningfully identified.

This interpretivist orientation leads directly to an abductive reasoning approach, which moves iteratively between established governance theory and empirical documentary evidence, identifying the most plausible explanatory account of observed governance patterns. Rather than testing a pre-formed hypothesis or building theory inductively from raw data, abductive logic uses COBIT 2019 and ISO/IEC 38500 as theoretical lenses through which DMCR's governance failures are interpreted, with the analysis refined iteratively as patterns emerge Ebneyamini (2022). These commitments are operationalized through a qualitative single-case study design, and Design Science Research provides the overarching methodological paradigm. These four elements form a coherent nested logic: interpretivism defines how organizational reality is understood; abductive reasoning defines how evidence is interpreted; the qualitative case study defines how the empirical work is structured; and DSR defines what the research is ultimately for — the design of a deployable governance artefact that directly resolves a documented organizational problem.

3.3 Research Methodology: Design Science Research

The DSR was formalized by Hevner et al. (2004) as a paradigm that generates prescriptive knowledge through the iterative design, development, and evaluation of artefacts intended to resolve identified organizational problems. This study adopts the six-stage DSRM process model of Peffers et al. (2007): (1) Problem Identification and Motivation; (2) Definition of Objectives for a Solution; (3) Design and Development; (4) Demonstration; (5) Evaluation; and (6) Communication. Within Gregor and Hevner's (2013) taxonomy, this constitutes an improvement contribution — a well-documented governance problem addressed through a novel, context-specific artefact set using established solution approaches applied to a new problem context, consistent with the precedent of Sengik et al. (2022).

3.4 Research Design: Qualitative Single-Case Study

The DSR methodology is operationalized within a qualitative single-case study design, adopting Yin's (2018) definition of a case study as an empirical inquiry that investigates a contemporary phenomenon in depth within its real-world context. DMCR constitutes a single bounded case, selected through purposive sampling based on its direct relevance to the cross-border governance deficiency typology under investigation, The availability of sufficient secondary documentary evidence, alongside prior authorized access to internal organizational documentation, supported the research. The single-case design is appropriate because the study's objective is analytical generalization rather than statistical generalization across multiple sites (Yin, 2018).

3.5 Secondary Data Sources and Collection

This study relies exclusively on secondary documentary data, with no involvement of human participants or primary data collection methods. This approach aligns with the requirements of a single-case Design Science Research (DSR) study and adheres to the approved ethics protocols as mentioned in the Appendix 2 (Project Reference P18478).

Five categories of documentary sources were utilized: (1) internal governance and financial policy documents; (2) financial records evidencing cross-border discrepancies; (3) project management artefacts, including communication and escalation logs; (4) audit and compliance reports; and (5) external regulatory frameworks, notably COBIT 2019 (ISACA, 2019) and (ISO/IEC, 2015). Internal documents constitute pre-existing organizational artefacts and were accessed with formal authorization. They are treated strictly as documentary evidence, with no analysis of individual perspectives. All sensitive data have been anonymized, and analysis is conducted at the governance-structure level.

Methodological triangulation is employed across three evidence streams: internal documents, industry benchmarking reports, and cross-sector case studies. Key external sources, including the Global Site Landscape Survey (SCRS, 2025), and GAP-CTS (WHO, 2025) which supports the generalizability of findings by demonstrating that DMCR's governance issues are sector-representative.

3.6 Data Analysis Procedure and Method-to-Analysis Traceability

Data analysis was conducted in three sequential phases, ensuring clear traceability between empirical findings and artefact design.

Phase 1 – Deficiency Identification (Chapter 4.1):

Documentary data were analyzed using thematic analysis (Braun & Clarke, 2006) to identify governance deficiencies across decision rights, accountability, escalation mechanisms, and financial controls. Cross-sector comparisons provided triangulated validation.

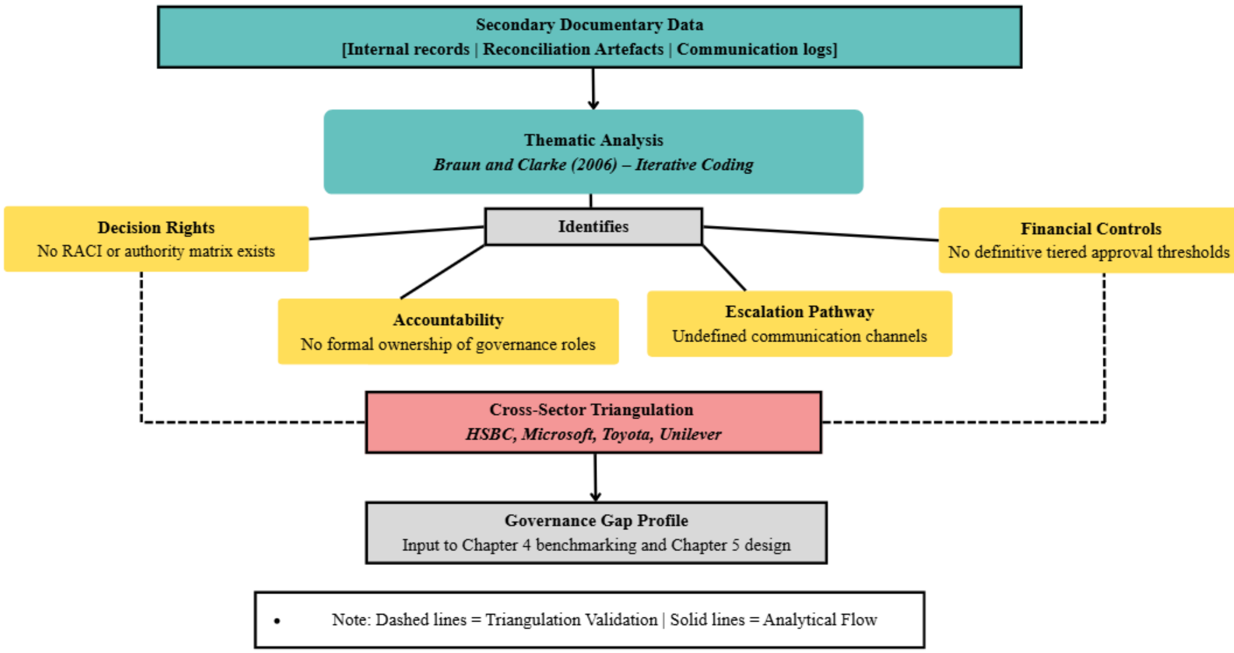

Figure 3.1: Thematic analysis and triangulation procedure. (Personal collection, 2026)

Phase 2 – Standards Benchmarking (Chapter 4.2-4.4):

Identified deficiencies were mapped against COBIT 2019 EDM objectives and ISO/IEC38500 principles, enabling structured gap analysis. Governance maturity was assessed using COBIT's five-level capability model (Toifur, et al., 2022), producing a matrix that links empirical findings to design requirements.

Phase 3 – Artefact Design (Chapter 5):

The artefacts were directly derived from the gap analysis, with each addressing specific, evidenced deficiencies. A clear problem-to-solution traceability is maintained, ensuring alignment with DSR principles (Hevner, et al., 2004).

3.7 Quality and Trustworthiness

Research quality is assessed using established trustworthiness criteria. (McDonald, et al., 2020). Credibility is ensured through triangulation and benchmarking against recognized governance standards. Transferability is supported by detailed contextualization of DMCR's environment. Dependability is maintained through a transparent, three-phase analytical process, providing an auditable trail (Lincoln & Guba, 1989). Confirmability is achieved by grounding interpretations in documentary evidence and established frameworks.

3.8 Ethical Consideration

A Certificate of Ethical Approval was obtained from the Arden University Ethics Board (Appendix 2, Project Reference P18478). As the study uses only secondary data, ethical considerations focus on confidentiality, responsible data handling, and accurate reporting. All organizational documents were accessed with formal premising and managed in accordance with institutional data protection policies. Sensitive information has been anonymized, and analysis is restricted to structural governance issues. No conflicts of interest are present.

Chapter 4 Results & Analysis

4.1 Organizational Governance Deficiency Analysis

Thematic analysis of DMCR's secondary documentary evidence, comprising internal financial reconciliation records, payment processing artefacts, inter-departmental communication logs, and operational policy documentations reveals four interconnected governance deficiency categories. Each is analyzed below in terms of its governance meaning, the documentary basis for that assessment, its operational and strategic consequences for DMCR, and its direct implication for the framework artefacts designed in Chapter 5.

Deficiency 1: Absence of formally defined decision rights.

The reviewed documentation contains no RACI-equivalent authority matrix, no governance charter, and no policy document allocating financial approval authority, discrepancy resolution ownership, or payment escalation responsibility between the U.S. and Pakistan functions. This indicates that, in governance terms, the absence of such artefacts reflects a structural gap in the core mechanism by which accountability is defined and assigned. The operational consequence is accountability diffusion, whereby financial discrepancies lack clear ownership, and ambiguity over responsibility itself becomes a source of delay De Smet et al. (2022). Unresolved discrepancies consequently accumulate in the absence of clearly defined authority, undermining audit trail integrity and sponsor confidence at DMCR. This deficiency directly informs the development of Artefact 1 in Chapter 5.

Deficiency 2: Absence of formal escalation protocols.

Inter-departmental communication logs document recurring instances of financial discrepancies that remained in informal email exchanges for extended periods without reaching a designated decision authority. No escalation protocol, trigger threshold, or resolution timeframe is documented in the reviewed policy or operational records. In terms of governance this represents the structural absence of the mechanism that transforms an individual-level operational problem into an organizationally managed governance event. Without a defined escalation pathway, resolution depends on individual initiative and interpersonal relationships rather than institutional obligation – a condition Wang, et al. (2025) identify as the primary driver of discrepancy accumulation and information asymmetry in cross-border governance contexts. The strategic consequence is that governance failures that could be resolved early instead become embedded problems across successive payment cycles. This deficiency directly motivates Artefact 4 in Chapter 5.

Deficiency 3: Weak financial oversight and approval controls.

The Financial reconciliation records show that transactions of materially different risk levels — from routine low-value payments to high-value sponsor reimbursements — are processed through the same undifferentiated approval pathway. No tiered approval thresholds, dual-authorization requirements, or exception-handling protocols are documented. In governance terms, absence of formally tiered approval thresholds means transactions of varying materiality are processed without commensurate oversight, this directly violates COBIT 2019's EDM03 Risk Optimization objective ISACA (2019), which requires that approval thresholds be defined commensurate with transaction risk, and ISO/IEC 38500's Conformance principal ISO/IEC (2015), which demands that governance practices be demonstrably compliant with applicable regulatory requirements. Operationally, this creates the conditions in which high-value or exceptional transactions pass through an approval process designed for routine items, generating both compliance exposure and audit risk. This deficiency directly motivates Artefact 2 in Chapter 5.

Deficiency 4: Structural communication failure between the U.S. and Pakistan teams.
Policy documentation, communication logs, and operational records collectively reveal the absence of any formal cross-border coordination mechanism between the U.S and Pakistan functions. No agreed reporting cadence, shared report template, governance checkpoint schedule, or information-sharing protocol was identified in the documentary record. In governance terms, this means the two functions operate without a shared information architecture – decisions in one function are regularly made without reliable knowledge of the current state in the other, producing the systematic information asymmetry that is visible throughout the reconciliation records. The strategic consequence is that financial reconciliation becomes reactive and error-prone, accurate cross-border financial positions cannot be produced on demand for sponsors or auditors, and the organization lacks the governance visibility required to detect emerging problems before they become material failures. This deficiency directly motivates Artefacts 3, 5, and 6 in Chapter 5.

4.1.1 Cross-Sector Governance Analogues

The primary findings above are grounded exclusively in DMCR's documentary evidence. Cross-sector analysis is included here not to substitute for that evidence but to provide triangulated validation that DMCR's deficiency profile conforms to established patterns of distributed governance failure – and that the intervention types required to resolve them are empirically validated across multiple organizational contexts. This is consistent with the triangulation methodology described in Section 3.5.

The HSBC governance fragmentation case U.S Senate, (2012) validates the pattern identified in Deficiencies 1 and 4: structural separation of operations across jurisdictions with undefined decision rights and absent coordination mechanisms produces systematic financial control failures regardless of individual competence. Toyota's escalation failure during the 2009–2010 accelerator recall crisis, Greto, et al., (2010) report validates the pattern in Deficiency 2: safety-critical signals remained unresolved in informal communication channels because escalation pathways were structurally undefined — a governance failure mode identical to DMCR's documented discrepancy accumulation. Microsoft's governance transformation Weill & Ross (2004) and Unilever's matrix governance experience Bartlett & Ghoshal (2002) together validate the principle underpinning Deficiencies 1 and 4: governance effectiveness in distributed organizations requires structural embeddedness — formal authority allocation and coordination mechanisms — not behavioral compliance. These analogues confirm that the artefact-based interventions designed in Chapter 5 address governance failure typologies with demonstrated effectiveness across sectors.

4.2 COBIT 2019 Benchmarking and Maturity Assessment

DMCR's governance arrangements were benchmarked against the five EDM domain objectives of COBIT 2019 using the five-level capability maturity model, based on the documentary evidence reviewed in Section 4.1 and consistent with the benchmarking methodology of Toifur et al. (2022). Table 4.1 presents the full benchmarking results.

COBIT 2019 EDM Obj.	Governance Obligation	DMCR Documentary Evidence	Maturity Level	Key Governance Gap Identified
EDM01	Ensure Governance Framework Setting and Maintenance	No documented governance framework or policy governing cross-border clinical-finance operations. Absence of any formal governance charter or authority matrix.	**Level 0** *Incomplete*	Complete absence of a formalized governance framework; no ownership of cross-border governance responsibilities assigned.
EDM02	Ensure Benefits Delivery	No evidence of governance value metrics or KPIs aligned to cross-border financial oversight objectives in available documentation.	**Level 1** *Initial*	No defined performance metrics linking governance activities to financial oversight outcomes; value delivery is ad hoc and unmeasured.
EDM03	Ensure Risk Optimization	No formal risk management protocol governing cross-border payment processing or financial discrepancy management. Risk responses are reactive and undocumented.	**Level 0** *Incomplete*	No risk register, risk appetite statement, or escalation trigger framework for cross-border financial operations.
EDM04	Ensure Resource Optimization	Finance team resourcing and task allocation between U.S. and Pakistan functions is undocumented; no formal RACI or role clarity framework exists.	**Level 1** *Initial*	Absence of a formal decision rights or RACI matrix allocating resource governance responsibilities across the two functional units.
EDM05	Ensure Stakeholder Engagement	Ad hoc email communication between U.S. clinical and Pakistan finance teams observed in records. No formal reporting cadence, escalation protocol, or governance checkpoint structure documented.	**Level 1** *Initial*	No formal cross-border communication governance mechanism; reporting is informal and inconsistent, producing information asymmetry between functions.

Table 4.1: COBIT 2019 EDM Domain Benchmarking — DMCR Governance Maturity Assessment

The benchmarking findings reveal a governance architecture that is structurally absent at its most critical points. EDM01 and EDM03 are both assessed at Level 0 — Incomplete. The documentary record was examined specifically for governance charters, authority matrices, risk registers, and risk appetite statements across both functions, and none were identified. The absence of these artefacts is the evidence for Level 0. EDM04 and EDM05 are assessed at Level 1 — Initial, reflecting ad hoc resource assignment and informal communication documented in operational records — activity that exists but is entirely unstructured and unmeasured. Collectively, this profile places DMCR significantly below the Level 3 (Established) threshold that Chaudhuri (2011) and ISACA (2019) identify as the minimum capability level for organizations governing complex, multi-function environments.

4.3 ISO/IEC 38500 Alignment Analysis

DMCR's governance practices were evaluated against all six principles of ISO/IEC 38500:2015, with alignment classified as Full, Partial, or Non-Aligned based on documentary evidence, consistent with the evaluation approach of Visitsilp and Bhumpenpein (2021). Table 4.2 presents the alignment findings.

ISO 38500 Principle	Standard Requirement	DMCR Current State (Documentary Evidence)	Alignment Status	Governance Implication for DMCR
Responsibility	All individuals and groups understand and accept their IT governance and decision-making responsibilities.	No documented decision rights allocation for cross-border financial governance. Neither the U.S. nor the Pakistan function has formally defined authority boundaries.	Non-Aligned	Critical gap: the absence of a RACI-based decision rights matrix means no role bears formal accountability for cross-border financial governance outcomes.
Strategy	IT plans satisfy current and future business requirements, incorporating governance and technology considerations.	No strategic governance plan exists for the integration of U.S. and Pakistan operational functions. Technology use is operational, not strategically governed.	Partial	Strategy is Partially Aligned: clinical trial strategy is well-defined but does not incorporate cross-border financial governance as a strategic priority.
Acquisition	IT acquisitions are made for valid reasons, with appropriate analysis and transparent decision-making.	Technology acquisition (including cloud analytics tools referenced in RQ4) lacks a formal governance evaluation or approval framework.	Partial	Governance of technology acquisition requires a formal approval threshold framework aligned to the financial oversight control layer designed in Chapter 5.
Performance	IT is fit for purpose in supporting the organization and providing services with required quality levels.	Clinical trial performance KPIs are documented; however, no performance metrics govern cross-border financial processing quality, timeliness, or discrepancy rates.	Partial	Financial governance KPIs and reporting transparency mechanisms must be designed to close the performance measurement gap in cross-border financial operations.
Conformance	IT complies with all mandatory legislation and regulations; policies and practices are clearly defined and enforced.	No formal compliance monitoring mechanism governing cross-border payment processing under U.S. clinical research regulations or applicable Pakistani financial regulations exists in reviewed documentation.	Non-Aligned	Critical gap: absence of a conformance framework creates regulatory exposure under FDA clinical research governance requirements and cross-border financial compliance obligations.
Human Behavior	IT policies, practices, and decisions demonstrate respect for human behavior, including the current and evolving needs of all people in the process.	Absence of formal policies addressing time-zone management, cross-jurisdictional communication norms, or cultural governance considerations between U.S. and Pakistan teams.	Non-Aligned	The distributed workforce governance gap is directly addressed by ISO/IEC 38500:2024 guidance on remote and geographically dispersed teams (ISO/IEC, 2024).

Table 4.2: ISO/IEC 38500:2015 — DMCR Governance Alignment Analysis.

The two most governance-critical findings are the Non-Aligned ratings against the Responsibility and Conformance principles. The Responsibility principle requires that authority for governance outcomes be explicitly allocated to defined roles (ISO/IEC, 2015). The documentary record demonstrates that no such allocation exists: no RACI matrix, no authority policy, and no formal role definition for cross-border financial governance was identified. This means that accountability for DMCR's cross-border financial governance is collectively assumed rather than individually owned — precisely the condition that produces the accountability diffusion documented in Deficiency 1. The Conformance principle requires that governance practices be demonstrably compliant with applicable regulations (ISO/IEC, 2015). No formal compliance monitoring mechanism for cross-border payment processing under either U.S. or Pakistani financial regulations was identified — creating measurable regulatory exposure on both sides of the operational structure.

4.4 Governance Gap Analysis Matrix:

Governance Deficiency (Evidenced)	COBIT 2019 EDM Objective	ISO 38500 Principle Violated	Chapter 5 Governance Artefact (Addressing Gap)
Absence of formally defined decision rights between U.S. clinical and Pakistan finance functions	EDM01 — Governance Framework Setting; EDM04 — Resource Optimization	Responsibility (Non-Aligned)	Artefact 1: Decision Rights Matrix — RACI model defining authority, accountability, consultation, and information rights across both functions
Undefined approval thresholds and weak financial oversight controls	EDM03 — Risk Optimization	Conformance (Non-Aligned); Performance (Partial)	Artefact 2: Financial Oversight Control Layers — tiered approval thresholds, exception handling protocols, and transaction governance controls
Absence of formal accountability allocation for reconciliation and reporting ownership	EDM04 — Resource Optimization; EDM02 — Benefits Delivery	Responsibility (Non-Aligned)	Artefact 3: Accountability Allocation Structures — formalized ownership of reconciliation, reporting, and exception handling roles
Undefined escalation protocols for cross-border financial discrepancy resolution	EDM05 — Stakeholder Engagement; EDM03 — Risk Optimization	Conformance (Non-Aligned); Human Behavior (Non-Aligned)	Artefact 4: Escalation Protocols — time-bound, role-assigned escalation pathways for discrepancy resolution and inter-function dispute management
Absence of governance review and compliance monitoring mechanisms	EDM01 — Governance Framework Maintenance; EDM02 — Benefits Delivery	Conformance (Non-Aligned); Performance (Partial)	Artefact 5: Governance Review Checkpoints — periodic compliance monitoring, audit trail maintenance, and governance health review cadence
Structural information asymmetry and communication failure between U.S. and Pakistan functions	EDM05 — Stakeholder Engagement	Human Behavior (Non-Aligned); Performance (Partial)	Artefact 6: Reporting Transparency Mechanisms — KPIs, standardized reporting templates, and cloud analytics as governance visibility infrastructure

Table 4.3: Consolidated Governance Gap Analysis Matrix — DMCR

4.5 Synthesis and Correlation of Findings

The analysis presented in this chapter establishes a consistent and mutually reinforcing picture of structural governance failure across all four deficiency dimensions. DMCR's COBIT 2019 maturity profile — predominantly Level 0 to Level 1 across the EDM domain — places it significantly below the Level 3 threshold for organizations of comparable complexity (ISACA, 2019). Four of the six ISO 38500 principles show Non-Aligned or Partially Aligned status, grounded in specific documentary evidence at each point. Industry benchmarking from SCRS (2025), WCG Clinical (2024), and WHO (2025) confirm that these conditions are sector-representative rather than organizationally exceptional.

The single most important analytical conclusion of this chapter is that DMCR's deficiencies are structural, not behavioral. They arise from the structural absence of governance architecture — the formal frameworks, authority allocations, coordination mechanisms, and accountability structures that a cross-border organization of DMCR's scale requires but does not currently possess. This distinction determines the nature of the required intervention: behavioral deficiencies call for training or personnel change; structural deficiencies call for governance architecture. Chapter 5 is designed to provide precisely that architecture.

The traceability between this chapter and Chapter 5 is exact and deliberate:

- **Artefact 1** responds to Deficiency 1 — the absence of formally defined decision rights.
- **Artefact 2** responds to Deficiency 3 — the absence of tiered financial oversight controls.
- **Artefact 3** responds to Deficiency 4 at the accountability ownership level.
- **Artefact 4** responds to Deficiency 2 — the absence of escalation protocols.
- **Artefact 5** responds to Deficiency 4 at the governance review level.
- **Artefact 6** responds to Deficiency 4 at the reporting and information transparency level.

No artefact in Chapter 5 is introduced without a traceable evidential foundation in this chapter, and no deficiency identified here is left without a direct governance intervention. This chapter clarifies the specific governance deficiencies at DMCR and establishes how each of the six subsequent artefacts constitutes a targeted, evidence-based structural response.

Chapter 5: Transformation Project Definition Framework Design

5.1 Framework Overview and Design Rationale

This chapter presents the governance-led decision support framework designed as the primary artefact of this study, in fulfilment of Research Objective O4 and in direct response to the governance deficiencies identified in Chapter 4. The framework comprises six interrelated artefacts, each derived from the gap analysis matrix in Table 4.3 and grounded in the COBIT 2019 EDM domain objectives and ISO/IEC 38500:2015 principles benchmarked in Chapter 4. The design logic is explicitly DSR-compliant: each artefact constitutes a prescriptive, deployable governance tool addressing a specific, evidenced deficiency in DMCR's cross-border governance arrangements (Hevner, et al., 2004). Weill and Ross (2004) establish that effective governance frameworks must specify who makes decisions, how decisions are made, and how decision-making performance is monitored — a tripartite architecture that the six artefacts collectively operationalize for DMCR's U.S.–Pakistan context.

5.2 Expected Project Benefits

Four categories of measurable benefit are expected, each justified by the Chapter 4 findings. First, governance maturity advancement: implementation is projected to elevate DMCR from its current Level 0–1 baseline to Level 3 (Established) across all five COBIT 2019 EDM objectives — the minimum threshold ISACA (2019) specifies for complex, multi-function organizations. Second, financial control and regulatory risk reduction: the tiered approval architecture (Artefact 2) and formally defined decision rights (Artefact 1) will eliminate the undifferentiated approval pathway evidenced in Chapter 4, reducing exposure to unauthorized transactions and FDA compliance risk. Third, operational efficiency: formally defined escalation protocols (Artefact 4) will replace the informal email-dependent discrepancy management process documented in Chapter 4, reducing resolution time consistent with Wang et al.'s (2025) findings on cross-border escalation effectiveness. Fourth, sponsor confidence: standardized reporting and governance checkpoints (Artefacts 5 and 6) will enable DMCR to produce a reliable, real-time cross-border financial governance picture on demand — a capability SCRS (2025) identifies as the primary determinant of sponsor trust at site level.

5.3 Alternative Business Options

Three alternatives were evaluated before selecting the framework approach:

- **Option 1**: Do noting – Is rejected on viability grounds: Chapter 4 establishes Level 0 maturity on DMCR's two most critical EDM objectives, making continued inaction inconsistent with DMCR's FDA-regulated obligations.

- **Option 2**: Personnel-based intervention – is rejected because adding staff without governance architecture distributes accountability diffusion rather than resolving it (De Smet, et al., 2022)

- **Option 3**: Third-party outsourcing – is rejected as unsustainable at DMCR's operational scale; it creates external dependency, produces no transferable institutional artefacts, and cannot deliver the standards-grounded framework the DSR methodology requires.

- **Option 4**: The internal DSR governance framework – is selected as the most viable, desirable, and achievable option because it directly and traceably resolves each structural deficiency, produces institutional capability owned by DMCR, and is phased to manage change incrementally within DMCR's existing operational structure.

5.4 Expected Project Disbenefits

Two significant disbenefits must be acknowledged. The primary drawback is short-term administrative burden: transitioning from informal to formal governance will impose additional obligations on both functions during implementation. Without proactive change management, this friction could slow transaction processing and produce partial compliance – formal adherence in process documentation without genuine behavioral change.

The secondary disbenefit is cultural resistance to formalization, particularly in the Pakistan finance function where informal relationships currently substitute for governance structures. If not managed with direct leadership sponsorship, resistance could result in superficial adoption of RACI matrix and escalation protocols, which is explicitly addressed in the risk register in Section 5.9.

5.5 Governance Artefacts

Artefact 1 – Cross-Border RACI Decision Rights Matrix

Artefact 1 addresses Deficiency 1 by assigning Responsible, Accountable, Consulted, and Informed designations across twelve financial decision categories. Authority is assigned at role level for structural durability across personnel changes, consistent with COBIT 2019 EDM04 (ISACA, 2019) and ISO/IEC 38500's Responsibility principle (ISO/IEC, 2015). The RACI designations are directly aligned with the project team roles in Section 5.7, linking governance design to project delivery accountability. Peterson (2004) establishes that RACI-structured frameworks are the most operationally effective tool for resolving accountability diffusion in distributed governance environments. [See Figure 5.1]

Artefact 1: Cross-Border RACI Decision Rights Matrix				
Decision Category	US Operations	Pakistan Finance Team	CFO	Sponsor/Audit
Payment initiation (Tier 1: Routine)	C	R	A	I
Payment approval (Tier 2: Mid-Range)	C	R	A	I
Payment approval (Tier 3: High Value)	C	R	A	I
Financial Discrepancy Identification	I	A	C	R
Discrepancy Escalation Trigger	R	A	I	C
Inter-Function Reconciliation	I	R	A	C
Sponsor Invoice Approval	C	R	A	I
Audit Trail Maintenance	I	C	R	R
Governance Review Participation	C	A	R	I

Legend: Responsible R - Accountable A - Consulted C - Informed I		
PK Finance = Pakistan-based finance function	US Ops = U.S. clinical operations	Authority assigned at role level

Figure 5.1: Artefact 1 Cross-Border RACI Decision Matrix. (Personal Collection, 2026)

Artefact 2 – Financial Oversight Control Layers

Artefact 2 addresses Deficiency 3 through three approval tiers: Tier 1 for routine transactions (single-function approval within a defined processing window); Tier 2 for mid-range transactions (dual-function approval by Pakistan Finance Officer and U.S. Clinical Operations lead); Tier 3 for high-value items (CFO authorization with mandatory audit trail entry). Each tier includes exception-handling protocols with automatic escalation triggers. Grounded in COBIT 2019 EDM03 and ISO/IEC 38500's Conformance and Performance principles (ISACA, 2019; ISO/IEC, 2015). Mikalef et al. (2021) demonstrate that tiered governance control architectures reduce financial processing errors and improve compliance audit outcomes in multi-site organizations. [See Figure 5.2]

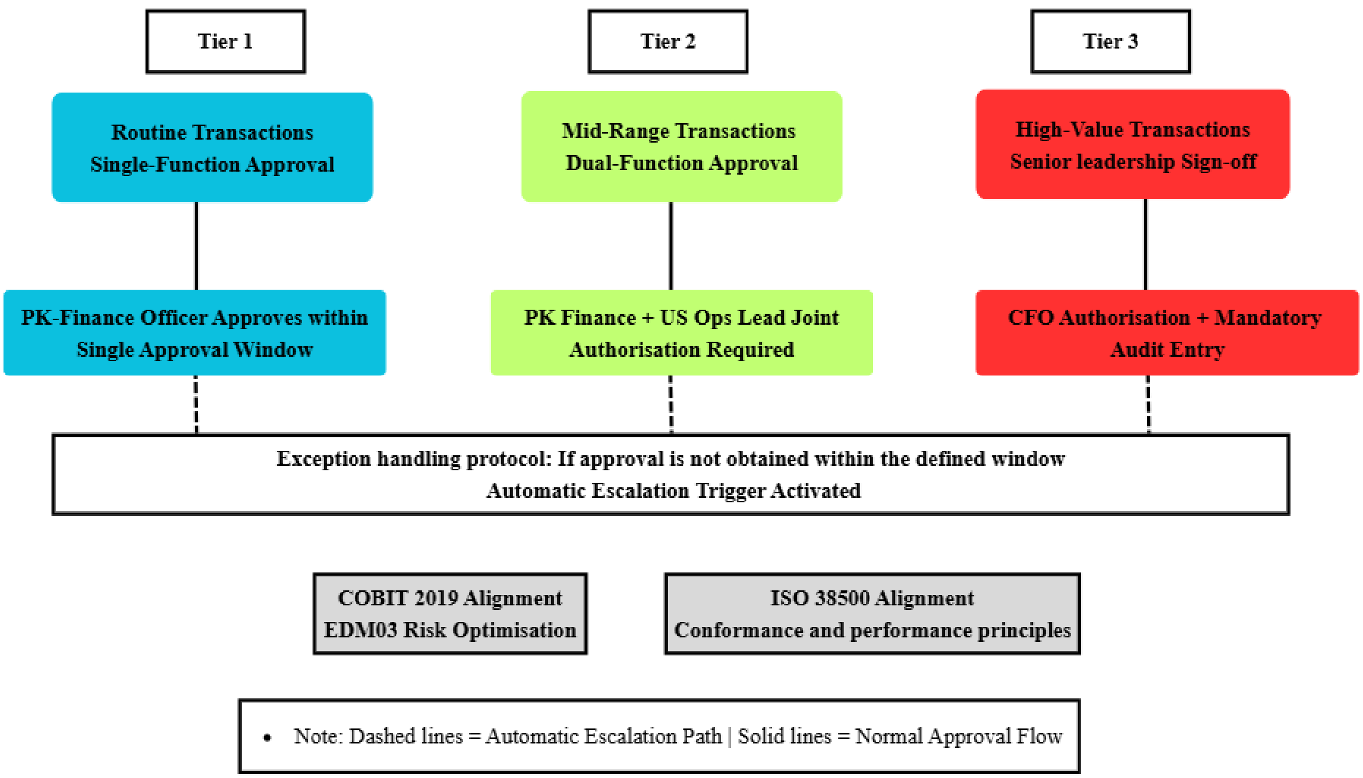

Figure 5.2: Artefact 2 Financial Oversight Control Layers. (Personal collection, 2026)

Artefact 3 – Accountability Allocation Structures

Artefact 3 addresses Deficiency 4 by assigning explicit ownership across six governance functions: monthly reconciliation; quarterly sponsor reporting; exception report management; audit trail custody; governance review participation; and escalation log management. Implements COBIT 2019 EDM04 and EDM02 and ISO/IEC 38500's Responsibility principle (ISACA, 2019). Sari (2023) demonstrates that formal accountability allocation is the governance intervention most strongly associated with improved financial oversight outcomes in comparable organizational settings. [See Figure 5.3]

Governance Function	Accountable (A)	Responsible (R)	Frequency
Cross-border Financial Reconciliation	PK Finance Manager	Finance Analyst	Monthly
Quarterly Sponsor Reporting	US Site Operations Director	PK Finance Manager	Quarterly
Exception Report Preparation and Circulation	PK Finance Manager	Finance Analyst	As Needed
Audit Trail Maintenance and Custody	CFO	Finance and Audit Teams	Continuous
Governance Health Review Participation	CFO	Finance and Site Teams	Quarterly

- Note: Each function carries a single accountable role; accountability is not collectively shared. ISO 38500: Responsibility principle | COBIT 2019: EDM04 Resource Optimisation + EDM02 Benefits Delivery |

Figure 5.3: Artefact 3 Accountability Allocation Structures. (Personal collection, 2026)

Artefact 4 – Escalation Protocols

Artefact 4 addresses Deficiency 2 through three escalation levels: Level 1 (48-hour intra-function resolution); Level 2 (72-hour cross-function joint resolution with mandatory log entry); Level 3 (CFO authorization with root-cause documentation triggered when Level 2 fails). Satisfies COBIT 2019 EDM05 and EDM03 and ISO/IEC 38500's Conformance and Human Behavior principles (ISACA, 2019; ISO/IEC, 2015). Wang et al. (2025) demonstrate that defined escalation pathways are the most effective structural intervention for reducing discrepancy accumulation in cross-border governance contexts. [See Figure 5.4]

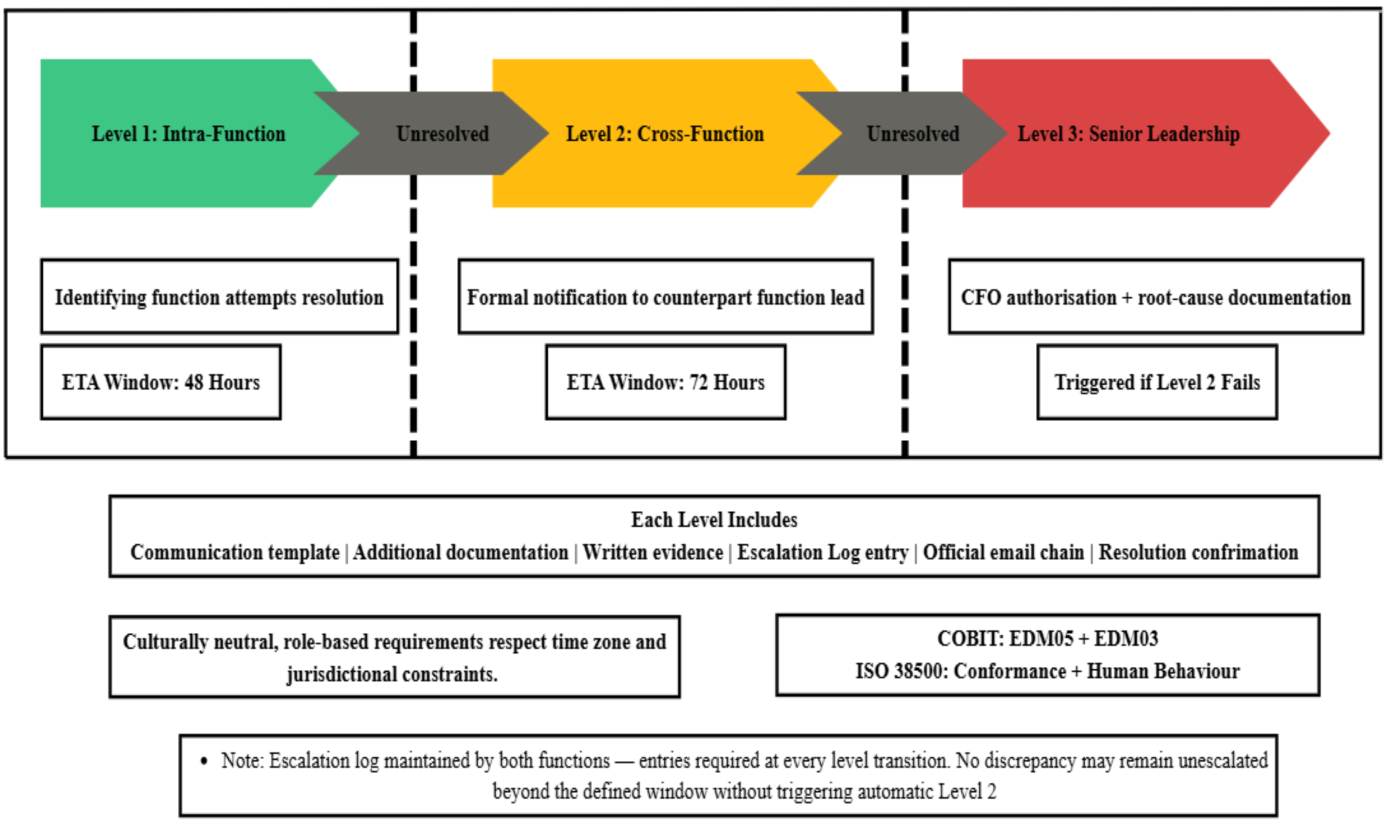

Figure 5.4: Artefact 4 Escalation Protocols. (Personal collection, 2026)

Artefact 5 – Governance Review Checkpoints

Artefact 5 addresses Deficiency 4 through three review cadences: Monthly Operational Reviews (transaction volumes, exception rates, escalation logs); Quarterly Compliance Reviews (COBIT 2019 and ISO/IEC 38500 performance, formal Governance Health Report circulated to senior leadership in both functions); Annual Governance Framework Reviews (fitness for purpose against operational and regulatory changes). Satisfies COBIT 2019 EDM01 and EDM02 and ISO/IEC 38500's Performance principle (ISACA, 2019). Müller et al. (2023) establish that periodic governance review mechanisms are the most effective structural tool for sustaining governance effectiveness in multi-site environments. [See Figure 5.5]

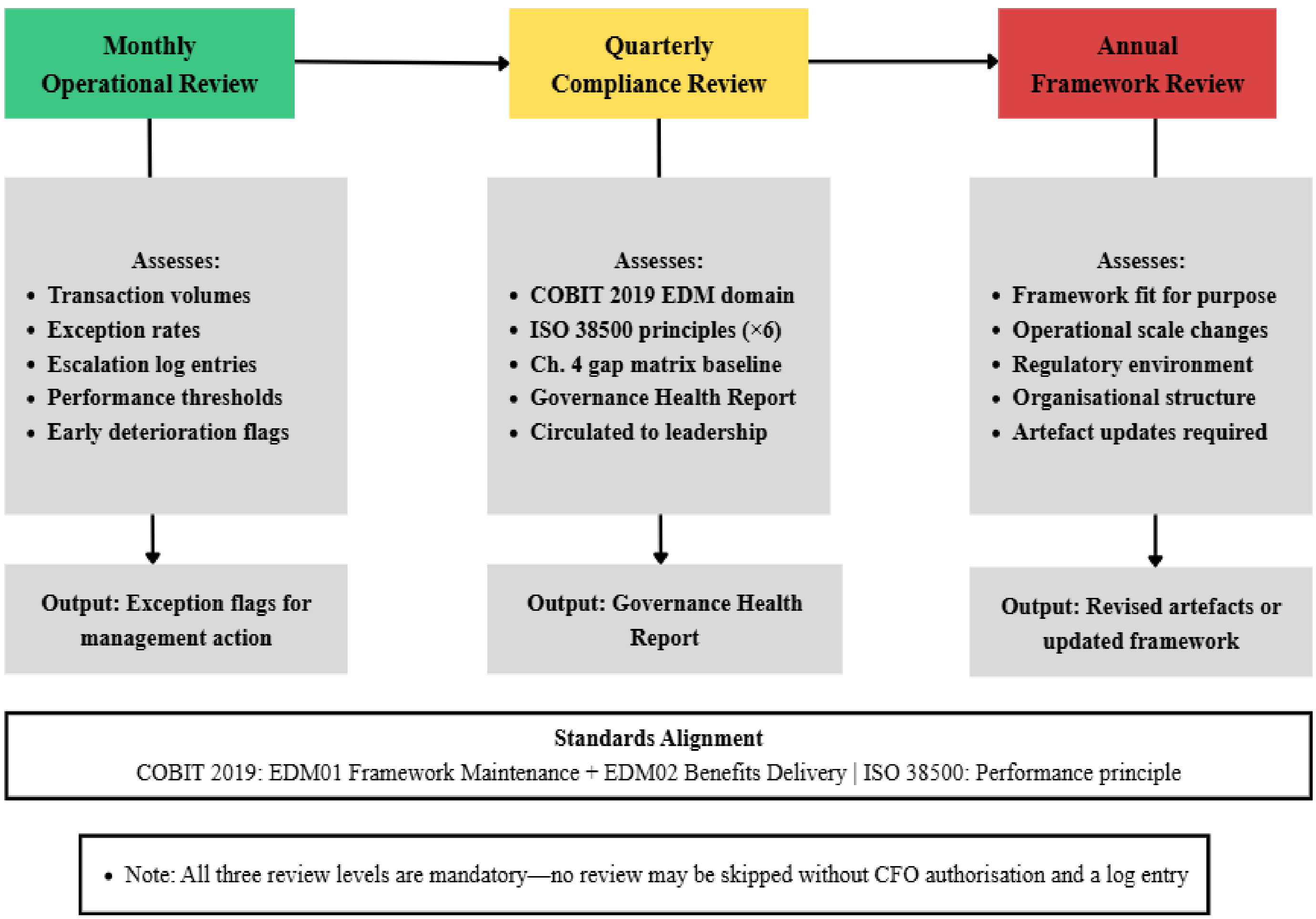

Figure 5.5: Artefact 5 Governance Review Checkpoints. (Personal collection, 2026)

Artefact 6 – Reporting Transparency Mechanisms

Addresses Deficiency 4 at the information level through standardized reporting templates, four defined KPIs (payments cycle time, discrepancy resolution rate, reconciliation accuracy, review completion rate), and Power BI dashboards as governance visibility infrastructure. Cloud analytics supports but does not substitute for Artefacts 1-5, consistent with (Abudaqqa, 2025). Satisfies EDM0, EDM02, and IS)/IEC 38500’5 Performance and Human Behavior principles. (ISACA, 2019); (ISO/IEC, 2015). [See Figure 5.6]

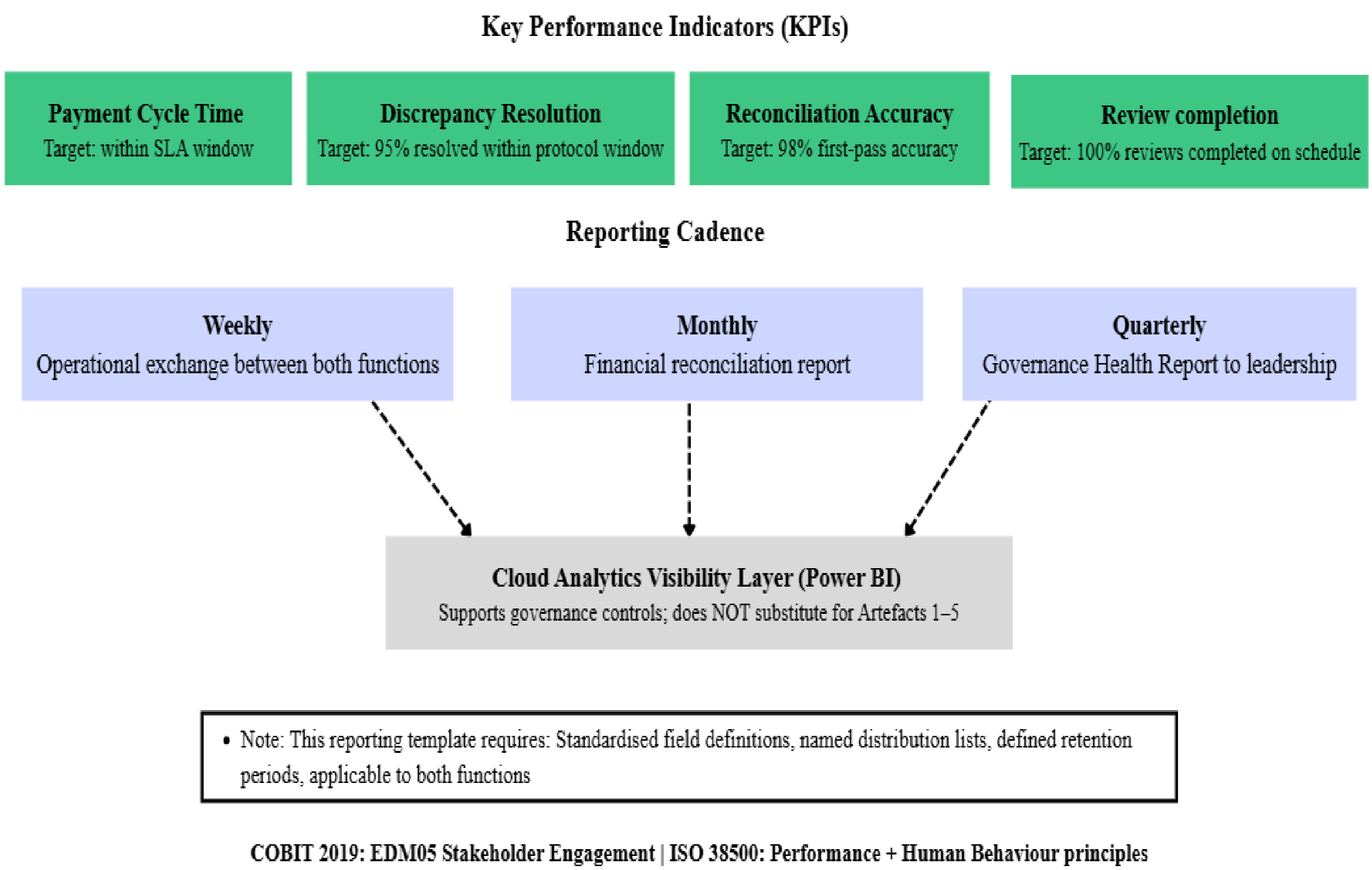

Figure 5.6: Artefact 6 Reporting Transparency Mechanisms. (Personal collection, 2026)

5.6 Phased Implementation Roadmap

The six artefacts are deployed across three sequential phases calibrated to DMCR's operational constraints. Phase 1 (Months 1–2: Foundational Governance) deploys Artefacts 1 and 3 simultaneously — neither is effective without the other. Phase gate: RACI formally signed, and all six governance function owners confirmed. Phase 2 (Months 3–4: Operational Controls) deploys Artefacts 2 and 4, which require Phase 1's authority structure to route approvals and escalations correctly. Phase gate: one full payment cycle processed under the three-tier structure without critical exception. Phase 3 (Months 5–6: Governance Sustainability) deploys Artefacts 5 and 6, operationalizing the monitoring and communication infrastructure to sustain Phases 1 and 2 controls. Phase gate: all KPIs baselined and first Quarterly Compliance Review scheduled. The Association for Project Management (2019) identifies phased governance implementation as best practice for organizations with limited change management capacity. [See Figure 5.7].

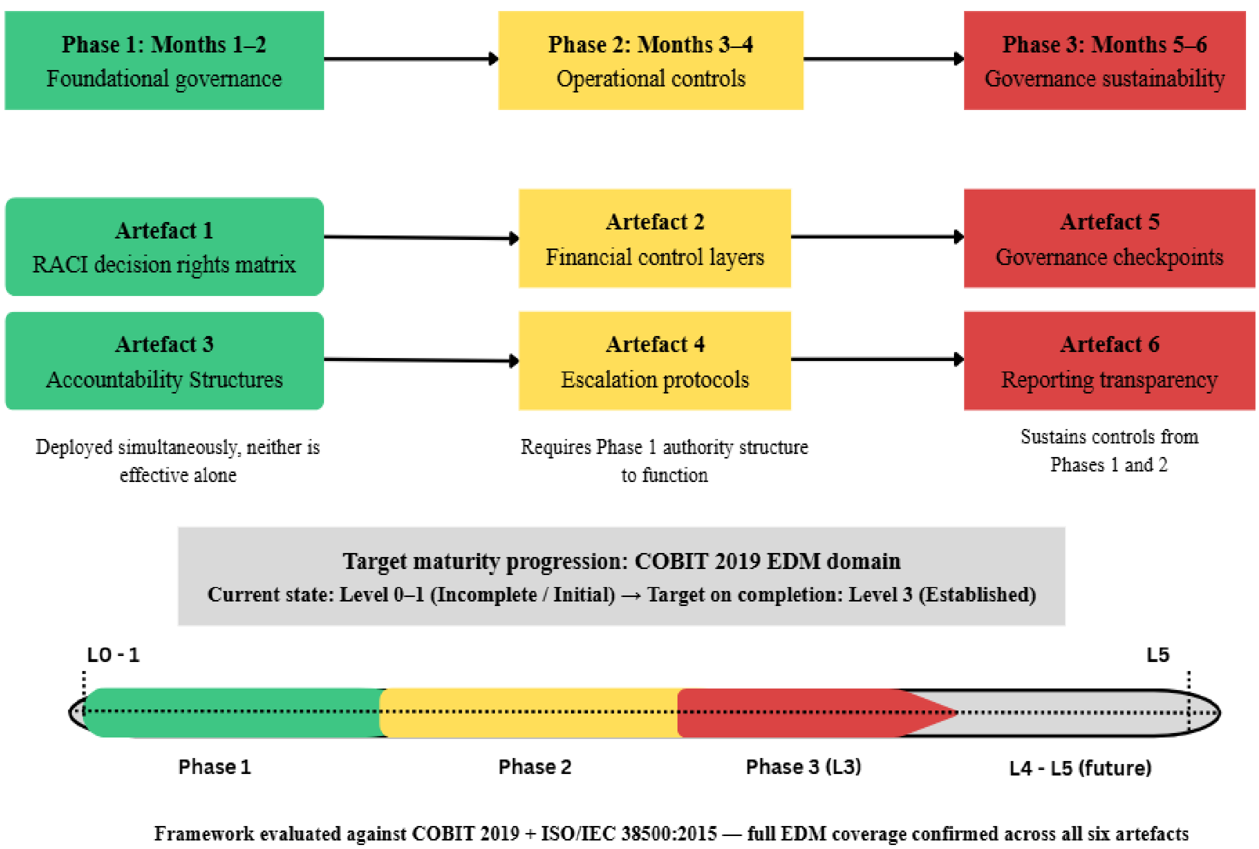

Figure 5.7: Chapter 5 Framework: Phased Implementation Roadmap for DMCR. (Personal collection, 2026)

5.7 Project Management Team Structure

The project team is distinct from the governance artefacts themselves. The Project Sponsor (CFO) holds accountability for overall transformation outcomes and phase gate sign-off. The Project Manager (PMO and Financial Compliance Administrator) holds responsibility for day-to-day delivery, stakeholder coordination, and artefact deployment. The U.S. Operations Lead (Senior Clinical Operations Director) represents U.S. function requirements and validates RACI assignments. The Pakistan Finance Lead (Senior Finance Manager) co-designs escalation protocols and accountability structures. An optional External Governance Adviser provides independent COBIT 2019 and ISO/IEC 38500 compliance validation during Phase 3. The project delivery RACI assigns the CFO as Accountable for all phase gate decisions, the Project Manager as Responsible for all delivery activities, and both function leads as Responsible for function-level artefact adoption — directly mirroring the role hierarchy in Artefact 1. [See Figure 5.8]

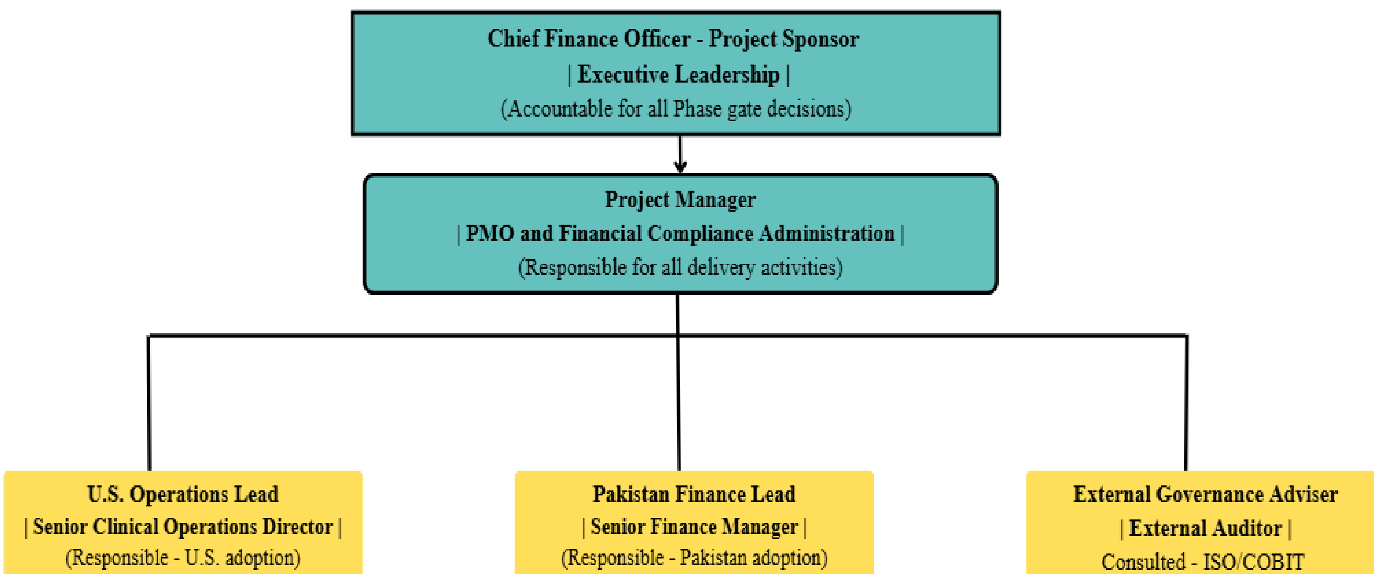

Figure 5.8: Proposed Project Management Team Structure for DMCR Governance Transformational Project. (Personal Collection, 2026)

5.8 Project Delivery Approach

A governance transformation at DMCR is best delivered through a hybrid project management approach, combining the structured sequential control of PRINCE2 (AXELOS, 2017) in Phase 1 with iterative agile sprint cycles in Phases 2 and 3. Phase 1 deploys the RACI matrix and accountability structures — artefacts that are structurally prerequisite to all subsequent governance controls and must be formally validated before any other artefact can operate correctly. PRINCE2's managed-stage model is directly applicable here, requiring each stage to be formally authorized before the next begins and providing the Project Sponsor with structured exception reporting throughout (AXELOS, 2017). Phases 2 and 3 introduce artefacts whose optimal configuration cannot be fully determined in advance — escalation time windows, KPI targets, and reporting templates require refinement from operational experience — making iterative sprint cycles the more appropriate delivery mechanism for these phases.

The Project Management Institute defines a hybrid approach as one that combines predictive and adaptive elements, selecting each based on the characteristics of the work rather than applying a single methodology uniformly (Project Management Institute, 2021). This principle applies directly to DMCR's transformation: Phase 1 work is predictive because its requirements are fully known, and outputs must be precisely validated; Phases 2 and 3 work is adaptive because it benefits from empirical feedback before final configuration. Across all three phases, PRINCE2's governance principles — defined roles, structured reporting, and a maintained business case — provide the program-level oversight appropriate for a cross-border, dual-function transformation of this kind, with formal CFO gate authorization at the close of Phase 1 and Phase 2 providing the control points at which continuation is approved or replanned. *[See Figure 5.9]*

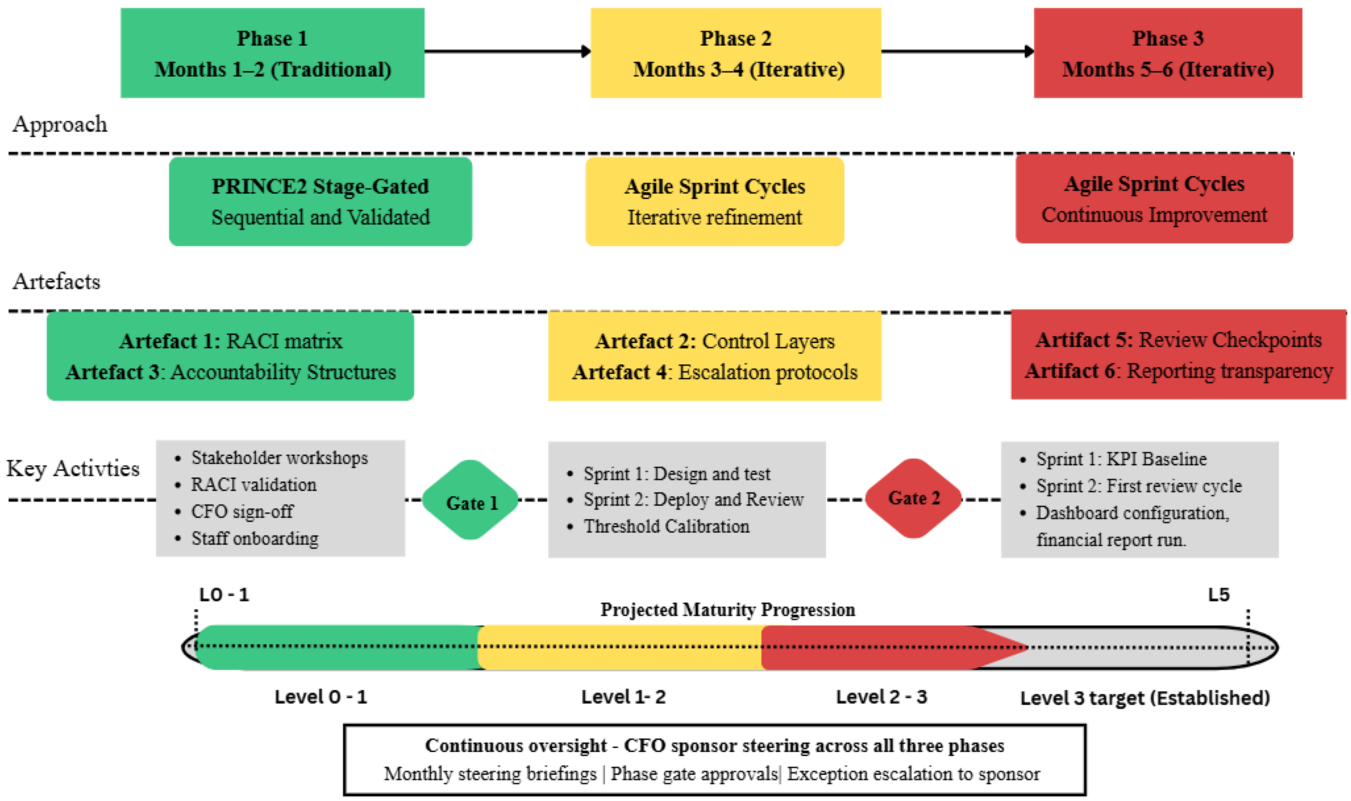

Figure 5.9: Proposed "Hybrid Project Delivery Approach" for DMCR Governance Transformation. (Personal collection, 2026)

5.9 Project Risk Register

Table 5.1 presents the four principal project risks with type, likelihood, impact, and mitigation measures

Risk I.D	Risk	Type	Likelihood	Impact	Mitigation
R01	Stakeholder resistance to formalisation, particularly in the Pakistan finance function	Strategic	Medium	High	CFO-sponsored change management programme; Pakistan Finance Lead co-designs artefacts to build ownership; phased rollout reduces cultural disruption
R02	Superficial adoption—formal process compliance without genuine behavioural change	Strategic	Medium	High	Monthly escalation log audits and quarterly Governance Health Reports (Artefact 5) detect real versus nominal adoption; CFO visibility creates accountability
R03	Transaction processing delays during Phase 1 and Phase 2 as staff adapt to new approval workflows	Operational	High	Medium	Tier 1 approval windows calibrated to match current informal processing times, running in parallel during Phase 1 transition
R04	Key personnel unavailability in either function during deployment	Operational	Medium	Medium	Role-level artefact design ensures continuity; RACI matrix assigns deputy responsibilities for all accountable roles

Table 5.1: Project Risk Register for DMCR Governance Transformation Plan. (Personal collection, 2026)

5.10 Stakeholder Engagement Plan

Table 5.2 presents the stakeholder engagement plan. The communication strategy is justified by DMCR's cross-border context: parallel engagement tracks for both functions prevent governance changes from being perceived as imposed by one function on the other; culturally neutral, role-based communication respects jurisdictional and cultural differences; and sponsor visibility of governance improvements directly delivers the reputational benefit identified in Section 5.2.

Stakeholder	Interest in Project	Engagement Approach	Frequency
CFO (Project Sponsor)	Governance maturity, regulatory compliance, financial risk reduction	Monthly steering briefings; phase gate sign-off, and receives Governance Health Reports	Monthly and at phase gates
U.S. Clinical Operations Director	Decision rights clarity, discrepancy burden reduction, audit trail reliability	Artefact design workshops; RACI validation; monthly operational review participation	Bi-weekly during deployment; monthly in Phase 3
Pakistan Finance Manager	Role clarity, protocol co-design, workload implications	Co-design workshops for Artefacts 3 and 4; KPI target-setting for Artefact 6	Weekly during Phases 1 and 2; bi-weekly in Phase 3
Pakistan Finance Analysts	Process changes, approval and reporting training	Governance onboarding training at Phase 1 and Phase 2 go-live; written SOPs distributed	At phase go-live; monthly updates
Sponsor Partners and Auditors	Audit trail integrity, payment governance, compliance assurance	Quarterly Governance Health Report circulated per sponsor agreement requirements	Quarterly

Table 5.2: Stakeholder Engagement Plan for DMCR Governance Transformation Project.

5.11 Project Cost Summary

Cost Category	Estimated Cost (USD)
Project Manager time—6 months, 0.5 FTE	$18,000
U.S. Operations Lead—workshops and reviews	$6,000
Pakistan Finance Lead—workshops and training	$4,000
COBIT 2019 and ISO/IEC 38500 licensing	$2,500
Power BI licensing and configuration (Artefact 6)	$3,600
Staff governance onboarding training — both functions	$5,000
External Governance Adviser — Phase 3 (optional)	$8,000
Contingency (10%)	$4,710
Total	$51,810

Table 5.3: DMCR Governance Transformation Project's Estimated Budget Summary. (Personal collection, 2026)

5.12 Framework Evaluation Against COBIT 2019 and ISO/IEC 38500:2015

The framework is evaluated using the standards-alignment method validated by Baskerville et al. (2018). Against COBIT 2019, the framework addresses all five EDM objectives: EDM01 through Artefacts 5 and 6; EDM02 through Artefacts 3 and 5; EDM03 through Artefacts 2 and 4; EDM04 through Artefacts 1 and 3; and EDM05 through Artefacts 4 and 6. Against ISO/IEC

38500:2015, all six principles are addressed: Responsibility (Artefacts 1 and 3); Strategy (Artefact 5); Acquisition (Artefact 2); Performance (Artefacts 5 and 6); Conformance (Artefacts 2 and 4); and Human behaviors (Artefacts 4 and 6). This confirms that the framework systematically resolves each deficiency in the Chapter 4 gap analysis matrix, with the measurable outcome being DMCR's elevation from Level 0–1 to the Level 3 (Established) threshold across the COBIT 2019 EDM domain (ISACA, 2019).

Standard Objective / Principle	Artefacts Addressing It
COBIT 2019 EDM01: Governance Framework Maintenance	**Artefacts 5 and 6**
COBIT 2019 EDM02: Benefits Delivery	**Artefacts 3 and 5**
COBIT 2019 EDM03: Risk Optimization	**Artefacts 2 and 4**
COBIT 2019 EDM04: Resource Optimization	**Artefacts 1 and 3**
COBIT 2019 EDM05: Stakeholder Engagement	**Artefacts 4 and 6**
ISO 38500: Responsibility	**Artefacts 1 and 3**
ISO 38500: Strategy	**Artefact 5**
ISO 38500: Acquisition	**Artefact 2**
ISO 38500: Performance	**Artefacts 5 and 6**
ISO 38500: Conformance	**Artefacts 2 and 4**
ISO 38500: Human Behavior	**Artefacts 4 and 6**

Table 5.4: Framework Evaluation Against COBIT 2019 and ISO/IEC 38500:2015. (Personal collection, 2026)

Chapter 6 Personal Reflection

This chapter reflects three key personal learnings from the Business Transformation Project process, using (Kolb, 1984) experiential learning cycle as a reflective framework. The reflection focuses on what I learned through carrying out the project, how it developed my thinking and capability, and how it will support my future professional practice.

6.1 Learning One: Distinguishing Structural Problems from Behavioral Ones

The most significant learning from this project was developing the ability to distinguish between structural governance failures and behavioral ones. Before undertaking this research, I tended to interpret governance problems at DMCR as issues of communication or individual working practices. The process of conducting thematic analysis against COBIT 2019 and ISO/IEC 38500 taught me to look beyond symptoms and identify the structural absence of governance architecture as the cause. This shift in analytical thinking has directly changed how I approach compliance challenges in my role as PMO and Financial Compliance Administrator. I now ask not whether people are behaving incorrectly, but whether the governance structures exist to make correct behavior institutionally possible. This is a capability I will carry into every governance endeavor that I will pursue in my professional career.

6.2 Learning two: Translating Frameworks into Deployable Practice

Working extensively with COBIT 2019 and ISO/IEC 38500 revealed that familiarity with governance frameworks is fundamentally different from the ability to operationalize them. While I began this project with prior exposure to both standards through my compliance role, the artefact design process required translating abstract governance principles into specific, role-level, operationally deployable tools capable of changing institutional behavior. Designing the RACI matrix, escalation protocols, and tiered control structures within real organizational constraints developed a practical implementation capability that extends far beyond theoretical knowledge. This applied competence is the most significant and transferable outcome for my future practice as a senior project management professional.

6.3 Learning three: Managing Research Rigor Under Professional Constraints

The third key learning was personal rather than technical. Undertaking rigorous research within my own organization required the development of strong analytical detachment. I was compelled to assess DMCR's governance deficiencies objectively, relying solely on verifiable documentary evidence rather than prior professional familiarity. This process strengthened my capacity for critical, evidence-based organizational analysis and has informed a more methodologically rigorous approach to how I evaluate and present governance findings in my professional practice going forward.

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Appendix 1: Approved Dissertation Project Title and Proposal

"Developing a Governance-Led Decision Support Framework to Strengthen Cross-Border Financial Oversight at DMCR." P18478

ARDEN UNIVERSITY

Medium Risk Research Ethics Approval

Project title

"Developing a Governance-Led Decision Support Framework to Strengthen Cross-Border Financial Oversight at DMCR."

Record of Approval

Principal Investigator

I request an ethics peer review and confirm that I have answered all relevant questions in this checklist honestly.	X
I I confirm that I will carry out the project in the ways described in this checklist. I will immediately suspend research and request new ethical approval if the project subsequently changes the information I have given in this checklist.	X
I confirm that I, and all members of my research team (if any), have read and agreed to abide by the Code of Research Ethics issued by the relevant national learned society.	X
I confirm that I, and all members of my research team (if any), have read and agreed to abide by the University's Research Ethics, Governance and Integrity Framework.	X
I understand that I cannot begin my research until this ethics application has been approved.	X

Name: Mr. Mirza Muhammad Wali Baig (24125373) (RES7005)

Date: 12/02/2026

Student's Supervisor (if applicable)

I have read this checklist and confirm that it covers all the ethical issues raised by this project fully and frankly.I also confirm that these issues have been discussed with the student and will continue to be reviewed in the course of supervision.

Name: Ahmad Al-Kaakaty

"Developing a Governance-Led Decision Support Framework to Strengthen Cross-Border Financial Oversight at DMCR." P18478

ARDEN UNIVERSITY

Dissertation Proposal Form

Student Name	Mr. Mirza Muhammad Wali Baig (24125373)	**Supervisor Name**	Ahmad Al-Kaakaty
Student Number	24125373	**Course**	MSC in MSc Project Management
Start and end dates of research	20 Feb 2026 - 20 Apr 2026		
Proposed activity – aims, objectives, research question(s), and state how it is novel			

Mirza Muhammad Wali Baig (24125373) (RES7005) Page 1 16 Feb 2026

"Developing a Governance-Led Decision Support Framework to Strengthen Cross-Border Financial Oversight at DMCR." P18478

1. Proposed Activity

1.1 Aim

To design a governance-led decision support framework that establishes financial oversight, accountability structures, and decision-making protocols to address discrepancies and communication failures between DMCR's global clinical and finance operations, with cloud-based analytics tools positioned strictly as supporting mechanisms for governance visibility.

1.2 Objectives

The following five objectives progress from diagnostic analysis through to framework design and evaluation. They are calibrated to a 9,000-word secondary-data case study. Cloud and AI tools are treated throughout as supporting mechanisms, not transformation drivers.

1.Objective 1: To analyze existing governance and financial oversight practices within DMCR using secondary organizational documentation, including internal policies, financial reporting records, audit findings, and process artifacts.
2.Objective 2: To benchmark DMCR's current governance practices against two core governance standards, COBIT 2019 (Governance and Management Objectives) and ISO 38500 (IT Governance Principles), to establish a structured evaluation baseline.
3.Objective 3: To identify governance gaps in decision rights allocation, accountability structures, escalation mechanisms, and financial control frameworks through structured gap analysis of documentary evidence.
4.Objective 4: To design a governance-led decision support framework addressing identified structural deficiencies, specifying decision rights, accountability allocation, escalation protocols, and financial oversight controls.

1.3 Research Questions

All research questions are calibrated to secondary-data analysis and design-based validation. No causal claims, experimental testing, or empirical proof is asserted.

5.RQ1 – Governance Deficiencies: What governance deficiencies contribute to financial discrepancies and communication failures between DMCR's global clinical and finance functions, as evidenced through secondary organizational documentation?
6.RQ2 – Standards Alignment: How do DMCR's current governance practices align with COBIT 2019 and ISO 38500 governance principles?
7.RQ3 – Framework Design: What governance structure can be designed to improve financial oversight, accountability clarity, and decision transparency across DMCR's cross-border operations?
8.RQ4 – Technology as Enabler: How can cloud-based analytics tools support (but not replace) governance controls and decision rights structures within DMCR's operational context?

1.4 Novelty of the Study

•Cross-Border Governance Gap: Governance frameworks exist for domestic clinical operations and multinational corporations separately. This study addresses a documented gap: governance design for cross-border clinical-finance integration where clinical sites operate in the U.S. and centralized finance operates in Pakistan. Existing literature treats these as discrete governance domains; this project integrates them.
•Governance-First Positioning: Unlike technology-centric implementations that layer tools onto broken processes, this project establishes governance primacy, defining decision rights, accountability, and control mechanisms first, then positioning cloud-based analytics as supporting visibility infrastructure.
•Time-Bounded Transformation Artefact: The project produces defined governance artefacts (decision rights matrix, accountability structure, escalation protocols, and control framework) as deliverables of a bounded transformation initiative, applying project governance theory to a finance-clinical integration context rarely addressed in existing scholarship.
•Practical Output with Theoretical Rigor: The framework bridges academic governance theory (COBIT 2019 and ISO 38500) with actionable implementation guidance, providing DMCR with deployable governance structures while contributing a replicable model for similar organizations.

Methodology – rationale, data selection and collection, recruitment, participant demographics, analytical process

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2. Methodology
2.1 Research Design
This project uses a qualitative case study design drawing exclusively on secondary organizational data and document analysis. No primary data collection, surveys, interviews, or human participant engagement will occur at any stage.

This project adopts a qualitative case study design (Yin, 2018) using DMCR as a single bounded organizational context. The study develops and evaluates a governance framework artefact through Design Science Research (DSR) principles (Hevner et al., 2004; Peffers et al., 2007), drawing exclusively on secondary data sources. DMCR serves as the design instantiation context: the organization within which the governance problem is situated and against which the artefact is designed and validated. It does not serve as a source of primary participant data.

2.2 Rationale for Methodological Choice
•Artefact-Creation Requirement: The primary output of this project is a governance framework to design an artifact, intended to resolve an identified organizational problem. A DSR approach within a case study context is the established methodology for creating and evaluating such artifacts grounded in theoretical rigor and practical utility.
•Secondary-Data Sufficiency: The governance failures under investigation, fragmented decision rights, absent escalation protocols, and weak financial controls are observable through documentary evidence, organizational policies, financial records, and published governance standards, without recourse to human participant data.
•Governance-Design Focus: The research questions are oriented towards framework design and standards-based evaluation, not towards lived experience or organizational perception, making secondary documentary analysis both methodologically appropriate and sufficient.

2.3 Data Sources
No primary data. No interviews. No surveys. All data sources are secondary and documentary in nature.

Data collection draws on five categories of secondary source:

•Internal Policies and Governance Documentation: Organizational governance policies, standard operating procedures, decision rights documentation, and internal communications protocols, accessed through formal organizational agreement with DMCR leadership and treated as artifact data.
•Financial Reporting Records: Annual financial reports (3–5 years), accounts receivable and payable records, reconciliation logs, and variance reports where available. These are treated as organizational artifacts, not participant data.
•Project Management Artifacts: Workflow documentation, reporting schedules, escalation pathway records, and operational process maps, used to map current-state governance structures.
•Audit and Compliance Documentation: Internal audit findings, compliance certifications, FDA inspection reports (publicly available via FDA.gov), and clinical trial registrations (ClinicalTrials.gov).
•Public Regulatory Standards Documentation: COBIT 2019 (ISACA), ISO/IEC 38500:2015, academic literature accessed via Scopus, Web of Science, ProQuest Business, EBSCO, and the Arden University digital library (2015–2025, emphasis on 2019–2025).

2.4 Analytical Techniques
1. Governance Benchmarking
Comparative analysis against:
- COBIT 2019
- ISO 38500

2. Governance Gap Analysis
Identification of deficiencies in:
- Decision rights
- Accountability structures
- Escalation mechanisms
- Financial oversight controls

3. Design Science (Single Iterative Cycle)
- Problem identification
- Framework design

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- Standards-based validation

2.5 DSR Framework Design Process (Single Iterative Cycle)
The governance framework is developed through a single DSR iterative cycle comprising five stages:

Stage 1: Problem Identification: Governance deficits identified through documentary analysis of DMCR secondary sources and gap analysis against COBIT 2019 and ISO 38500.

Stage 2: Objectives Definition: Governance outcomes required to resolve identified deficits, aligned to the five project objectives.

Stage 3: Design and Development: Iterative construction of framework components (decision rights matrix, accountability structure, escalation protocols, financial controls).

Stage 4: Demonstration: Application and instantiation of the framework within DMCR's cross-border operational context using documentary evidence.

Stage 5: Evaluation: Standards-based assessment of framework alignment with COBIT 2019 and ISO 38500.

2.6 Governance Framework Artefact Description
The artefact is a Project Governance Framework — not a technical system. Cloud-based analytics tools are described strictly as supporting mechanisms for monitoring, reporting, and analytical visibility, not as governance substitutes.

The framework will deliver the following governance artefacts as bounded transformation deliverables:

•Defined Decision Rights Matrix: A RACI model specifying financial decision authority across DMCR's U.S.–Pakistan clinical-finance operations.
•Financial Oversight Control Layers: Approval thresholds, exception-handling procedures, and financial control checkpoints aligned to governance standards.
•Accountability Allocation Structures: Clear ownership assignments for reconciliation, reporting, and escalation responsibilities.
•Escalation Protocols: Defined pathways for discrepancy resolution and time-zone coordination between U.S. clinical and Pakistan finance teams.
•Governance Review Checkpoints: Periodic oversight mechanisms to ensure ongoing framework compliance and performance monitoring.
•Reporting Transparency Mechanisms: KPI definitions, audit trail requirements, and oversight specifications. Cloud-based analytics tools are identified only as supporting infrastructure for these reporting functions.

2.7 Framework Evaluation Strategy
Evaluation is design-based, not empirical. No experimental testing, causal proof, or performance measurement is claimed. Standards alignment is the primary validation mechanism.

The framework will be evaluated through four structured assessment activities:

•Standards Alignment Assessment: Each framework component is mapped against COBIT 2019 governance objectives and ISO 38500 governance principles to verify structural compliance.
•Governance Completeness Analysis: Assessment of whether the framework addresses all identified governance gaps (decision rights, accountability, escalation, financial control) with no structural omissions.
•Logical Coherence and Control Coverage Review: Internal consistency check ensuring framework components interact coherently and controls provide adequate oversight coverage.
•Comparative Benchmarking: Comparison of the framework's governance design against recognized governance principles and published best practices from COBIT 2019 and ISO 38500 documentation.

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Project management (project timeline and key outputs)

Start date	Activity
04/02/2026	Governance & Mobilization (Week 1–2)
04/02/2026	Validation & Closeout (Week 20–22)
04/02/2026	Stakeholder Engagement & Change (Week 10–20)

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04/02/2026	Framework Definition & Playbooks (Week 14–20)
05/02/2026	Cloud/Data Architecture (Week 10–16)
05/02/2026	Analytics & AI Prototyping (Week 8–14)
05/02/2026	Survey & Data (Week 4–10)
05/02/2026	Requirements & Analysis (Week 2–6)
05/02/2026	Governance & Mobilization (Week 1–2)
05/02/2026	Overview: PID Approval, Scope, Outcomes, and Success Criteria

Research Data Management Plan (Describe the data you expect to acquire or generate during this research project, how you will manage, describe, analyse, and store the data and what mechanisms you will use to share and preserve your data.)

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3. Research Rigour and Quality
3.1 Trustworthiness Criteria
Research quality is assessed against Lincoln and Guba's (1985) trustworthiness criteria, adapted for a secondary-data qualitative case study context:

•Credibility: Triangulation across multiple secondary source types (academic literature, governance standards, industry documentation, and organizational artifacts) ensures findings are grounded in convergent evidence.
•Transferability: The framework is analytically generalizable through standards alignment, not statistical generalization. Detailed contextual documentation of DMCR's governance environment enables adaptation to similar cross-border clinical research organizations.
•Dependability: A comprehensive audit trail documents all data sources, analytical decisions, and framework design iterations, ensuring methodological transparency throughout.
•Confirmability: Framework design decisions are explicitly linked to documentary evidence. External validation criteria (COBIT 2019 and ISO 38500) reduce reliance on researcher interpretation and strengthen objectivity.

3.2 DSR Quality Criteria
The project is additionally assessed against Hevner et al.'s (2004) DSR quality principles:

•Design as an Artifact: A clearly bounded, documented governance framework constitutes the primary project output.
•Problem Relevance: Governance failures addressed are real, organizationally significant, and evidenced through documentary sources.
•Design Evaluation: Systematic assessment against COBIT 2019 and ISO 38500 standards.
•Research Rigor: Systematic secondary data collection and theoretically grounded design methods.
•Communication of Research: Final output is accessible to both academic and practitioner audiences, consistent with the BTP consultancy-oriented format.

4. Ethical Considerations
Ethics Declaration: This project involves NO human participants. All data is collected from secondary sources only. No surveys, interviews, focus groups, or human interaction of any kind will occur at any stage of this project.

4.1 Ethics Statement
This project is classified as secondary-data research with no human participant involvement. All data relies on published academic literature, publicly available governance standards, regulatory documentation, and organisational documentary artefacts treated as non-participant data. No personally identifiable information will be collected, processed, or analysed at any stage.

4.2 Ethical Safeguards
•Data Privacy: All internal organisational data will be anonymised and presented in aggregate form. No personally identifiable information will appear in the project report or supporting documentation.
•Confidentiality: Commercially sensitive operational information will be generalized or redacted in published outputs. DMCR's proprietary processes will be described at the governance-structure level only.
•Organizational Access: Access to internal documentary artifacts will be obtained through formal written agreement with DMCR leadership. Documents are treated as organizational artifacts, not participant-generated data.
•Intellectual Property: All published sources will be properly attributed in accordance with AU Harvard referencing conventions. Organizational proprietary information will be protected and not reproduced in publicly accessible outputs.

Planned outputs/publications/research datasets/impact/dissemination

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6. Planned Outputs, Impact, and Dissemination
6.1 Governance Artefacts (Primary Deliverables)
This project delivers a defined set of governance artifacts as the outputs of a time-bounded transformation initiative:

14.Governance-Led Decision Support Framework: A standards-aligned governance framework addressing cross-border financial oversight in clinical research operations, grounded in COBIT 2019 and ISO 38500 principles.
15.Decision Rights Matrix (RACI): Specification of financial decision authority across DMCR's U.S.–Pakistan clinical-finance operations.
16.Accountability and Escalation Protocol: Clear ownership structures for reconciliation, reporting, and discrepancy escalation with defined timelines and thresholds.
17.Financial Oversight Control Framework: Approval thresholds, exception-handling procedures, and governance control checkpoints.
18.Reporting Transparency Specifications: Measurable KPIs, audit trail requirements, and oversight mechanisms. Cloud-based analytics tools are identified as supporting infrastructure only.
19.Implementation Roadmap: A time-bounded plan detailing governance rollout sequencing, responsibilities, and resource implications for DMCR.

6.2 Research Datasets
This project uses secondary data only. No raw proprietary datasets or personally identifiable information will be published. Derived analytical outputs (gap analysis tables, standards alignment matrices) will be retained as part of the project audit trail. All organizational data is anonymized and aggregated.

6.3 Impact and Dissemination
•Organizational Impact (DMCR): Improved financial oversight, reconciliation accuracy, and escalation clarity across distributed operations; strengthened regulatory compliance posture; and a reusable governance blueprint for future operational scaling.
•Academic Impact: Contribution to applied governance literature through a replicable design artefact addressing cross-border clinical-finance integration, demonstrating the practical application of DSR in business transformation contexts.
•Dissemination: Submission of the final BTP report in partial fulfillment of degree requirements. Potential adaptation into a practitioner-oriented article (subject to institutional and organizational approval). Internal presentation to DMCR leadership where permitted.

If successful, I undertake to carry out the research according to the University's Ethics code of Practice and will be required to complete an Ethics checklist – see relevant forms detailed under Primary Research (Applicant's signature required)	12/02/2026 - Mr. Mirza Muhammad Wali Baig (24125373)
Date and signature of Supervisor approval	Date: 12/02/2026 - Ahmad Al-Kaakaty

Appendix 2: Certificate of Ethical Approval

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ARDEN UNIVERSITY

Certificate of Ethical Approval

Applicant:	Mr. Mirza Muhammad Wali Baig (24125373)
Project Title:	"Developing a Governance-Led Decision Support Framework to Strengthen Cross-Border Financial Oversight at DMCR."

This is to certify that the above named applicant has completed the Arden Research Management System Ethical Approval process and their project has been confirmed and approved as Medium Risk

Date of approval:	16 Feb 2026
Project Reference Number:	P18478

Glossary

APM	Association for Project Management
AXELOS	AXELOS Limited (owner of PRINCE2 and other best practice frameworks)
BAI	Build, Acquire, and Implement (COBIT 2019 domain)
BTP	Business Transformation Project
CFO	Chief Financial Officer
CIO	Chief Information Officer
COBIT	Control Objectives for Information and Related Technologies
COBIT 2019	Control Objectives for Information and Related Technologies (2019 edition)
DMCR	DM Clinical Research
DSR	Design Science Research
DSRM	Design Science Research Methodology
DSS	Delivery, Service, and Support (COBIT 2019 domain)
EDM	Evaluate, Direct, and Monitor (COBIT 2019 governance domain)
EDM01	Ensure Governance Framework Setting and Maintenance
EDM02	Ensure Benefits Delivery
EDM03	Ensure Risk Optimization
EDM04	Ensure Resource Optimization

EDM05	Ensure Stakeholder Engagement
FDA	U.S. Food and Drug Administration
FTE	Full-Time Equivalent
GAP-CTS	Global Action Plan for Clinical Trial Ecosystem Strengthening (WHO)
GCP	Good Clinical Practice
HSBC	Hongkong and Shanghai Banking Corporation
ICH	International Council for Harmonization
ICH-GCP	International Council for Harmonization — Good Clinical Practice
ISACA	Information Systems Audit and Control Association
ISO	International Organization for Standardization
ISO/IEC	International Organization for Standardization / International Electrotechnical Commission
ISO/IEC 38500	International Standard for IT Governance of Organizations
IT	Information Technology
KPI	Key Performance Indicator
MEA	Monitor, Evaluate, and Assess (COBIT 2019 domain)
MSc	Master of Science
O1–O4	Research Objectives 1 through 4
PK	Pakistan (used in tables to denote Pakistan-based finance function)

PMI	Project Management Institute
PMBOK	Project Management Body of Knowledge
PMO	Project Management Office
PRINCE2	Projects in Controlled Environments (version 2)
RACI	Responsible, Accountable, Consulted, Informed
RES7005	Module code for the MSc Business Transformation Project at Arden University
RQ1–RQ4	Research Questions 1 through 4
SCRS	Society for Clinical Research Sites
SLA	Service Level Agreement
SOP	Standard Operating Procedure
U.S.	United States
USD	United States Dollar
WCG	Western Compliance Group (WCG Clinical)
WHO	World Health Organization